

Berry curvature and response functions from VASP wavefunctions

VASPBERRY 1.6.1 - terminology revision

28 September 2026

Software edition: VASPBERRY 1.6.1. Terminology revision: 28 September 2026. Saved reference tables retain the producer identities and input checksums of their original calculation; software validation and material convergence are recorded separately.

Abstract

VASPBERRY calculates Berry curvature, topological indices and response properties directly from VASP wavefunctions. Its main workflow reads a WAVECAR with the native Fortran executable and writes numerical results for post-processing and plotting. The examples compare occupied-space and spin-sector Chern numbers evaluated with the Fukui-Hatsugai-Suzuki (FHS) method, the Z_2 invariant evaluated with the Fukui-Hatsugai (FH) method, and Berry curvature from the Kubo formula on Brillouin-zone meshes and symmetry paths. A native projected-spin Kubo task also resolves the two spin sectors, including the change of their basis with $\mathbf{k}$; its canonical-momentum approximation remains explicit. Native interband matrix elements also support chemical-potential and temperature scans through Python occupation weighting and integration. Circular optical selection and real-space spinor densities complete the direct-wavefunction demonstrations.

Monolayer and bilayer MoS_2 , graphene, a Bi bilayer and a three-septuple-layer MnBi_2Te_4 film illustrate these capabilities and their numerical limits. VASP supplies the electronic structure; its band energies provide context for VASPBERRY's topology and response results. Integer-valued topological invariants and Kubo-formula response integrals have different convergence requirements: an integer invariant does not establish a converged transport integral. Optional physical-operator extensions and VASP-derived Wannier checks are documented separately in the appendices. The report documents software capabilities and numerical limitations with reproducible inputs, commands and reference results.

1. Methods

1.1 Berry curvature from wavefunction overlaps

The Fukui-Hatsugai-Suzuki (FHS) method evaluates the Berry phase around each cell of a periodic $\mathbf{k}$ mesh using overlaps between neighboring wavefunctions. For an isolated group of bands, the overlap is a matrix and its determinant gives a gauge-invariant loop phase. With the connection $\mathbf{A} = i\langle u | \nabla_{\mathbf{k}} u \rangle$ used here,

$$\Phi_p = -\arg(U_1(\mathbf{k})U_2(\mathbf{k} + \Delta\mathbf{k}_1)U_1(\mathbf{k} + \Delta\mathbf{k}_2)^{-1}U_2(\mathbf{k})^{-1}), \quad \overline{\Omega}_p = \frac{\Phi_p}{\Delta S_k}.$$

The normalized link U_j is the phase of the overlap determinant. $\Delta S_k = |\mathbf{b}_1 \times \mathbf{b}_2|/(N_1N_2)$ is the area of a mesh cell, with reciprocal vectors containing 2π . Thus Φ_p is a phase and $\overline{\Omega}_p$ is an area-averaged curvature in $\text{\AA}^2$. The Chern number is $C = (2\pi)^{-1} \sum_p \Phi_p$. An isolated occupied group can include internal degeneracies; its separation from excluded bands remains essential. The discrete construction and its band-group extension are described by Fukui, Hatsugai and Suzuki.

The map and integral convey different information. A material can have strong local curvature at opposite valleys while its total Chern number is zero. The FHS link construction supplies a lattice Chern number; the Kubo integral is not rounded or adjusted to reproduce that integer. An integer lattice Chern number also needs a sufficiently resolved mesh and a physically appropriate isolated band group.

1.1.1 Projected-spin sectors

For an isolated selected band space, the native `spin-chern` task constructs the matrix of $\sigma_{\hat{a}}$ along a chosen Cartesian axis. It retains all off-diagonal band elements. Its positive and negative eigenspaces define the spin sectors, following Prodan. The FHS method evaluates their Chern numbers C_+ and C_- ; the spin Chern number is their half-difference:

$$C_{\text{charge}} = C_+ + C_-, \quad \Delta C = C_+ - C_-, \quad C_{\text{spin}} = \frac{C_+ - C_-}{2}.$$

Both the occupied-to-excluded energy separation and the projected-spin gap about zero must remain open; internal degeneracies within either subspace are allowed. Reversing the spin axis exchanges the sector labels. The WAVECAR-only calculation uses its full Gram matrix O and solves $\Sigma_{\hat{a}} v = \lambda O v$, with $v^\dagger O v = 1$, to form sector links in one consistent pseudo metric. It therefore declares a pseudo-wavefunction geometric representation, without PAW augmentation. The matching OUTCAR provides the spin-frame transformation; the wavefunctions and automatic component detection come from WAVECAR. The native guide specifies inputs, diagnostics and output tables. These geometric invariants are distinct from PROCAR-weighted charge attribution and the conventional spin-current response in Appendix A.2. A full occupied space describes the insulating system; an isolated subset describes only that subset. Their Chern numbers need not coincide.

1.2 Native Kubo curvature and Hall post-processing

For an isolated band and a Cartesian Hamiltonian derivative $D_\alpha = \partial H / \partial k_\alpha$,

$$\Omega_{n,xy}(\mathbf{k}) = -2 \text{Im} \sum_{m \neq n} \frac{D_{x,nm} D_{y,mn}}{(E_n - E_m)^2}.$$

This sum-over-states expression follows from the Kubo formula and evaluates Berry curvature at each sampled $\mathbf{k}$ point; see Xiao, Chang and Niu. Sharp curvature near small gaps makes the $\mathbf{k}$ sampling and intermediate-band window particularly important. Individual-band values are not used where the sampled bands cannot be resolved separately. The native WAVECAR implementation uses canonical momentum of the stored pseudo-wavefunctions; full material velocity can require PAW, nonlocal and SOC terms. The exported-matrix interface permits an explicitly supplied physical operator (Appendix A).

The ordinary WAVECAR route is the starting point for the charge-Hall examples and requires no VASP source modification. Two optional comparisons extend the operator description: standard VASP optical output for insulating occupied bundles, and an instrumented producer for full velocity and spin matrices. These input routes and their distinct scopes are summarized in the operator guide. A more complete operator addresses an operator approximation; its $\mathbf{k}$ -mesh and band-window convergence must still be tested.

For an isolated occupied bundle $\mathcal{V}$, the trace curvature is

$$\Omega_{xy}^{\mathcal{V}} = -2 \text{Im} \sum_{n \in \mathcal{V}} \sum_{m \notin \mathcal{V}} \frac{D_{x,nm} D_{y,mn}}{(E_n - E_m)^2}.$$

Internal occupied-pair terms cancel in the sum of band curvatures. Excluding them before division avoids singular individual-band terms at internal degeneracies. The bundle must remain separated from the excluded bands. This occupied-to-unoccupied formulation and its numerical advantage are given by Wang et al., Eq. (11) and Sec. III D.

For a two-dimensional system, the intrinsic charge sheet conductivity is

$$\sigma_{xy}(\mu, T) = -\frac{e^2}{\hbar} \sum_n \int_{\text{BZ}} \frac{d^2 k}{(2\pi)^2} f(E_{n\mathbf{k}} - \mu, T) \Omega_{n,xy}(\mathbf{k}).$$

Within an insulating gap at zero temperature, this becomes $\sigma_{xy} = -(e^2/h)C_{\text{occ}}$ in the stated convention. General background on Berry curvature and electronic transport is given in Xiao, Chang and Niu. The Bi Hall example evaluates the occupied-group flux directly; it does not assign independent curvatures to unresolved Kramers partners.

For chemical potentials crossing band edges, the Kubo implementation uses unordered interband pairs. With $N_{nm}^{xy} = -2 \text{Im}(D_{x,nm} D_{y,mn})$, the occupation-weighted integrand is $\sum_{n < m} (f_n - f_m) N_{nm}^{xy} / (E_n - E_m)^2$. Equal-occupation pairs cancel before division. Changes from a gap reference are evaluated with occupation differences before summation. This preserves small doping responses without subtracting large filled-band baselines. The temperature enters only the Fermi function; the electronic structure remains fixed. Regional contributions integrate the same charge response over specified parts of the BZ, with a $\mathbf{K}$ - $\mathbf{K}'$ difference defined without a factor of one-half. Such a partition is not a separately conserved valley-current operator.

1.2.1 Kubo curvature of projected-spin sectors

The native spin-kubo task evaluates the positive/negative sectors of Section 1.1.1 at each supplied k point. Simply rotating ordinary band Kubo curvatures into a spin basis misses the k dependence of that basis. The native calculation instead differentiates the projected-spin split within a declared local tangent approximation.

Let C contain the selected raw pseudo-wavefunctions, $O = C^\dagger C$, and $\Sigma = C^\dagger \sigma_a C$. Solve $\Sigma X = O X \Lambda$, $X^\dagger O X = I$, and form $U = C X$. The source WAVECAR supplies canonical-momentum matrix elements and the following approximate tangent, using retained states outside the group:

$$D_{mn}^a = \frac{\hbar^2}{m_e} \langle \tilde{u}_m | k_a + G_a | \tilde{u}_n \rangle, \quad \widetilde{\partial_a C_n} = \sum_{m \notin P} \tilde{C}_m \frac{D_{mn}^a}{E_n - E_m}.$$

By default all stored source bands contribute. `--sum-bands N` retains bands 1 through N outside the `--bands` interval, testing the response cutoff on unchanged VASP eigenstates. The cap must contain the selected interval and at least one external state; cutting an unresolved external degeneracy is an error. Parent isolation still checks all stored bands. Metadata records the source count, actual cap and boundary gap; states absent from WAVECAR remain untested.

No energy denominator between selected states is used. Define the horizontal tangent and the projected-spin derivative in the spin eigenframe:

$$B_a = (I - U U^\dagger) \widetilde{\partial_a C} X, \quad K_a = B_a^\dagger \sigma_a U + U^\dagger \sigma_a B_a.$$

For either sign sector s , its trace curvature is

$$\Omega_{ab}^s = -2 \operatorname{Im} \sum_{\alpha \in s} \left[(B_a^\dagger B_b)_{\alpha\alpha} + \sum_{\beta \in \text{opposite sector}} \frac{(K_a)_{\beta\alpha}^* (K_b)_{\beta\alpha}}{(\lambda_\alpha - \lambda_\beta)^2} \right].$$

The first and second contributions are exported separately. The second uses only opposite-sign spin eigenvalues; degeneracies within a sector remain allowed. It cancels between the two sector traces, whose sum equals the parent tangent curvature. A selected group that cuts an unresolved energy degeneracy is rejected. Closing of the projected-spin gap, invalid metric conditioning and nonfinite results are also errors.

The geometric kernel is exact for a correct supplied tangent. Its production canonical tangent is not the exact derivative of the stored pseudo projector: raw pseudo states are not a complete orthonormal physical PAW basis, and nonlocal/SOC velocity terms, PAW augmentation, basis derivatives and missing source bands are not restored by selected-space Gram orthonormalization. The output explicitly records these limits. This is a geometric spin-sector curvature approximation, not conventional spin-current Hall conductivity.

SPIN_KUBO.csv contains Cartesian yz, zx, xy curvature in $\text{\AA}^2$, source k coordinates, path distance, both contributions and gap/metric diagnostics. SPIN_KUBO_SPECTRUM.csv retains the projected-Pauli spectrum. Only an explicit `--mesh NX,NY` adds SPIN_KUBO_INTEGRAL.csv, with raw dimensionless integrals

$$C_s^{\text{est}} = \frac{1}{2\pi N_x N_y} \sum_{\mathbf{k}} (\Omega_{yz}^s, \Omega_{zx}^s, \Omega_{xy}^s) \cdot (\mathbf{b}_1 \times \mathbf{b}_2).$$

The signed area preserves the orientation for tilted cells. No path integration or integer rounding is performed. To determine a Chern number, run `spin-chern` separately on a complete mesh using the same band range and axis. The native guide, output contract and Sections 3.8–3.10 provide commands and real material comparisons.

Independent four-state tests with exact derivatives agree with small geometric sector loops within 6.94×10^{-9} in model units for all three components. Nonorthogonal selected-frame changes and axis reversal preserve the expected geometry. The exact-model positive-sector integral approaches the Chern number +1 obtained with the FHS method: 1.015414722 at 8×8 and 1.000000169 at 24×24 . The mixing term reaches 0.05207, so its omission is detectable. These tests validate the geometric kernel separately from material approximations.

1.3 From a WAVECAR to reusable results

The direct-wavefunction workflow has four steps:

1. Calculate the electronic states with VASP, using a full periodic mesh for BZ integrals or an explicit path for local curvature and band context.
2. Run the native VASPBERRY executable on the WAVECAR. Select the FHS method for Chern numbers, the FH n-field method for the Z_2 invariant, Kubo-formula calculations for pointwise curvature and charge response, or the optical and wavefunction output modes.
3. Retain the original numerical samples. For charge transport, the Python pair processor applies occupations, temperature and BZ or regional weights to native matrix elements; this is part of the response calculation. The common Python interfaces provide CSV, DAT and NPZ results where supported, with units and represented bands recorded.
4. Plot those outputs beside the material's VASP band energies. Display interpolation may smooth a map, but does not replace the sampled data used for integration or convergence checks.

The current repository build provides the following `--task` selectors:

Native task	Numerical output
chern	Berry flux, area-averaged curvature and Chern number of the selected band space, evaluated with the FHS method
z2	Z_2 invariant from the Fukui-Hatsugai (FH) n-field method
spin-chern	Spin-sector Berry fluxes and Chern numbers from the FHS method; spin Chern number (available in v1.6.1)
spin-kubo	Projected-spin sector point curvature and optional raw mesh integrals (available in v1.6.1)
kubo	Pointwise band or bundle curvature on a mesh or path
kubo-pairs	Interband-pair data for Python charge-Hall integration
optical / spectrum	Circular transition selectivity / broadened spectra
wavefunction	Selected Γ -state amplitudes for density plots

For example, `build/vaspberry --task kubo` selects the native pointwise calculation; supply the WAVECAR and band options listed in the usage guide and worked examples. The two projected-spin tasks are available in v1.6.1. Other task aliases are available in v1.4.0 and later releases; legacy short flags remain supported. The hands-on guide and report-to-example map give the input files, commands and expected outputs for each example.

Python prepares inputs, normalizes outputs and performs the stated post-processing. For `spin-chern` and `spin-kubo`, the native executable already writes the computed values; the example Python plotter only draws them. Optional standard-WAVEDER, supplied physical-matrix and Wannier interfaces use additional Python numerical backends with input requirements described in Appendix A and Appendix B.

1.4 Layer, orbital and spin character from PROCAR

The `procar_character.py` postprocessor connects a matching SOC PROCAR to the same final WAVECAR states and uses the actual spin-frame rotation printed in OUTCAR. User-defined atom groups describe layers or sublayers; optional orbital groups refine the character. For the validated noncollinear projection convention, write the raw charge and Pauli projections of group L as $q_L(\mathbf{n}\mathbf{k})$ and $\mathbf{m}_L(\mathbf{n}\mathbf{k})$. Along a specified Cartesian unit spin axis $\hat{\mathbf{a}}$, the joint projected weights are

$$p_{L,\pm}(\mathbf{n}\mathbf{k}) = \frac{q_L(\mathbf{n}\mathbf{k}) \pm \hat{\mathbf{a}} \cdot \mathbf{m}_L(\mathbf{n}\mathbf{k})}{2}.$$

For isolated selected bands, the saved native pair data give point curvature with the same canonical-momentum convention as Section 1.2. Multiplying by these weights defines a projected charge-Hall attribution:

$$\bar{\sigma}_{xy}^{L,\pm}(\mu, T) = -\frac{e^2}{\hbar} \sum_{n \in \mathcal{B}} \int_{\text{BZ}} \frac{d^2k}{(2\pi)^2} f(E_{n\mathbf{k}}; \mu, T) p_{L,\pm}(\mathbf{n}\mathbf{k}) \Omega_{n,z}(\mathbf{k}).$$

Here $\mathcal{B}$ is the explicitly selected band window. Its unweighted selected-band response is saved alongside the attribution; it is not the full occupied-state Hall response unless that window represents all relevant states. Raw projection coverage and residuals remain visible rather than being forced to unity. Selected states with unresolved degeneracies are rejected because individual band weights and curvature can depend on the chosen basis there.

This analysis supports layer/spin-colored band and curvature plots and comparisons of doping-induced projected contributions. Conventional spin Hall uses a spin-current operator and is treated separately in Appendix A.2; diagonal PROCAR character alone does not supply that operator. The public projection tutorial gives `project`, `hall` and `plot` commands with a small analytic fixture. That fixture tests software conventions and reproducibility; it is not a new VASP material calculation or a material-convergence benchmark.

2. Materials and numerical settings

Dataset	Wavefunctions and sampling	Quantities shown
Monolayer MoS ₂ , full zone	SOC; 12×12 mesh; 26 bands; 400 eV	Occupied-space Berry curvature from the FHS method and the Kubo formula, bands 1–18
Monolayer MoS ₂ , matching path	SOC; 49 K–Γ–K′ points; 26 bands; 400 eV	VASP band structure and Kubo curvature beside the maps
Monolayer MoS ₂ , local valleys	SOC; two 9×9 patches; 26 bands; 400 eV	Isolated band-18 curvature near K and K′
Monolayer MoS ₂ , transport	SOC; 12×12 to 36×36 meshes; stored and retained band windows varied separately; 400 eV	Chemical-potential and temperature dependence of regional Hall response
Monolayer MoS ₂ , operator comparison	Same 12×12 SOC eigenstates; 60 stored bands; pairs within 1–40 or 1–50; 400 eV	Ordinary WAVECAR and optional full PAW velocity Hall curves
MoS ₂ stacking comparison	Monolayer and 1H, 2H, 3R bilayers; PBE+SOC; separate 6×6 SCF densities; 49 Γ–M–K–Γ–K′ points; 60/64 path bands; 400 eV	VASP bands and circular optical selection from WAVECAR and standard WAVEDER
Monolayer MoS ₂ , supplied path	SOC; 48 K–Γ–K′ points; 32 bands; 400 eV	Γ-point state density; original optical tutorial
Buckled Bi bilayer	Two atoms; PBE+SOC; 400 eV; fresh 12×12 SCF density; full 6×6, 12×12 and 18×18 meshes; occupied bands 1–10	Native Z ₂ n-field; optional PAW spin response in Appendix A
MnBi ₂ Te ₄ , three septuple layers	SOC+U; 21 atoms; full 6×6 VASP mesh; 192 bands; occupied bands 1–123; 270 eV	Occupied-space Chern number from WAVECAR using the FHS method; coarse optical-integral diagnostic
Planar graphene	Two C atoms; PBE+SOC; 520 eV; 16 bands; occupied bands 1–8; full 6×6, 9×9 and 12×12 meshes	Native projected-spin sectors and Chern numbers; separate no-SOC gap control
Graphene spin-sector Kubo	Same fixed density; 49 Γ–K–M–Γ points and 12×12 mesh; 16 bands	Bands 1–8 and 7–8 local curvature; occupied-space Chern-number comparison; rejected pair link
Bi spin-sector Kubo	Same fresh Bi fixed density; 49 Γ–K–M–Γ points and 6×6 mesh; 48 bands; 400 eV	Bands 1–10 and 9–10 local curvature; different occupied/pair Chern numbers

The input guide links the structures and VASP files. The MoS₂ maps and their matching path use the same public SCF charge density, structure, potentials, cutoff and 26-band window. VASP 5.4.4 generated the 12×12 mesh and 49-point path with ICHARG=11 and ISYM=−1; only the k-point list changes. The sampled occupied-to-empty direct gap is 1.674 eV. The tutorial supplies preparation files and commands for both WAVECARs. The stacking examples use fixed, documented idealized structures and a common 24 Å cell height. The 3R slab additionally uses a self-consistent z dipole correction. These small path calculations illustrate optical analysis across input systems; no structural or full-zone response convergence is asserted. The stacking guide provides the geometries and complete VASP workflow. The real-space example and original optical tutorial retain the separate supplied 32-band path.

The optional Bi and MnBi₂Te₄ Wannier representations and their separate interpolation checks are described in Appendix B. Their model construction is additional to the WAVECAR workflows in the main text.

3. Results from VASP wavefunctions

3.1 MoS₂: Berry curvature from the FHS method over the full Brillouin zone

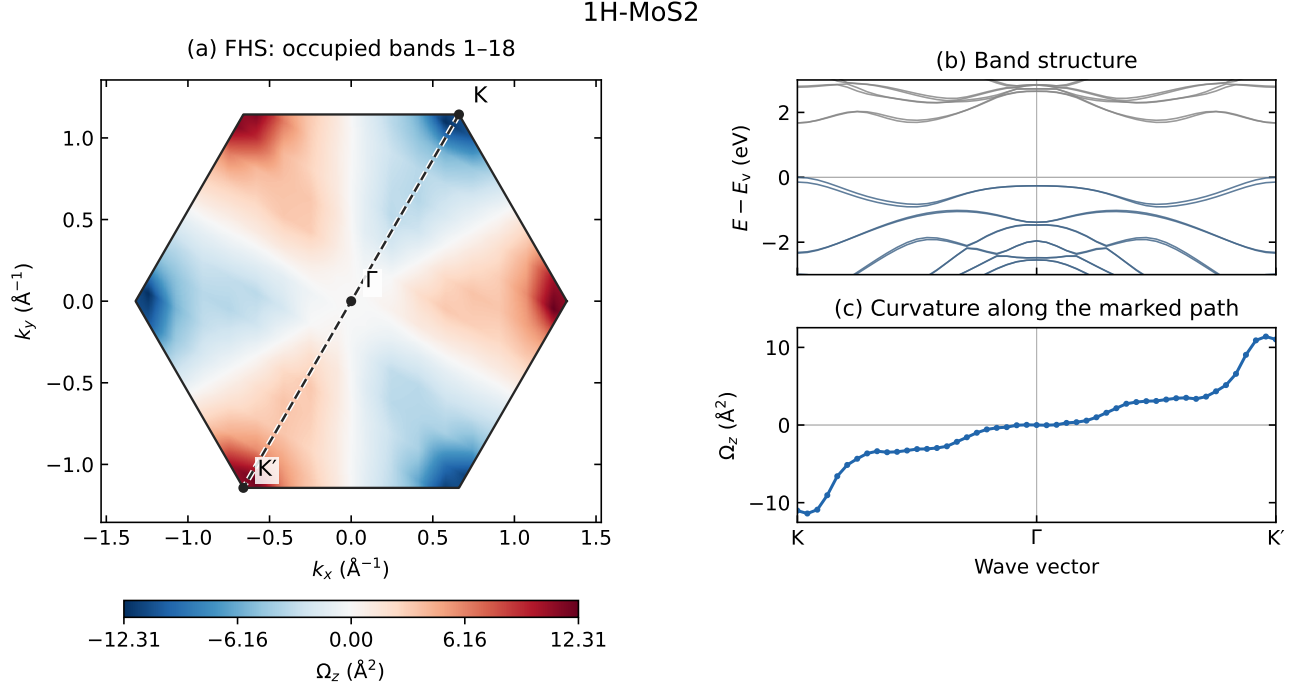


Figure 1. (a) Area-averaged Berry curvature from the FHS method of occupied bands 1–18 in the Cartesian first Brillouin zone. The dashed line marks K – Γ – K' , with $K = (1/3, 2/3)$ and $K' = -K$ in reciprocal coordinates. Native 12×12 plaquette values are displayed with periodic bilinear interpolation onto a 401×401 Cartesian grid. (b) VASP band structure along that path; energies are relative to the valence-band maximum, with occupied bands in blue and empty bands in gray. (c) A periodic bilinear cut of the plaquette field along the marked path. This line visualizes panel (a), rather than adding an independent pointwise curvature calculation. Both right-hand panels share the same Cartesian path distance.

The curvature has opposite signs near the time-reversed K and K' valleys. The largest sampled magnitudes are approximately $12.31 \text{ \AA}^2$, while the full-zone Chern number is zero to numerical precision. The native text map integrates to approximately -5×10^{-7} after decimal rounding. This illustrates the distinction between valley-contrasting local geometry and vanishing total charge Chern number. The connection between inversion breaking, spin–orbit coupling and valley optical response in MoS₂ is discussed by Xiao et al.. The present map is a finite-mesh example rather than a convergence study.

Calculation and plotting commands

3.2 MoS₂: Kubo curvature of the occupied valence-band bundle

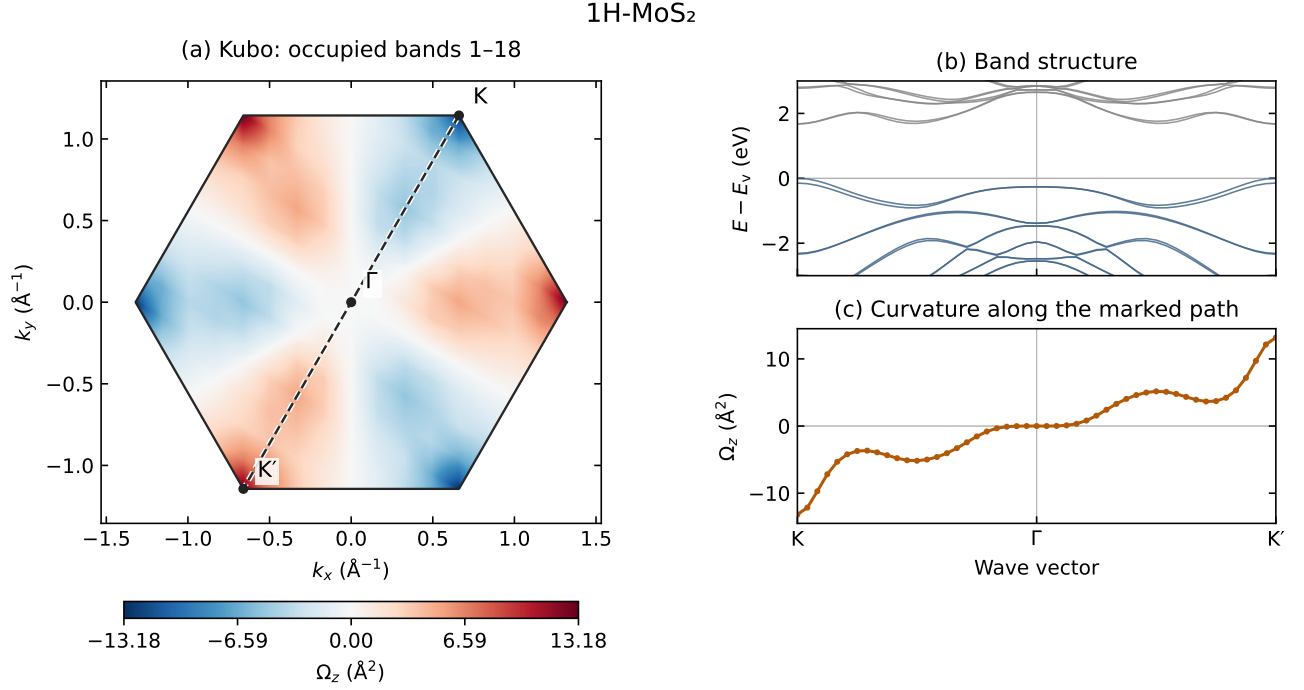


Figure 2. (a) Canonical-momentum Kubo trace curvature of occupied bands 1–18 on the 12×12 mesh. Only transitions to the stored empty bands 19–26 enter the sum. The map uses the same periodic bilinear display interpolation and dashed K – Γ – K' path as Figure 1. (b) VASP band structure, with occupied bands in blue and empty bands in gray. (c) Bundle curvature calculated directly at all 49 path points. Internal valence-band degeneracies do not interrupt the bundle curve.

The occupied-bundle curvature is approximately $-13.18 \text{ \AA}^2$ at K and $+13.18 \text{ \AA}^2$ at K' . The occupied space stays separated from the empty states by at least **1.674 eV** on both samplings, so every mesh and path point is valid for the bundle calculation. The native option `-kubo_bundle 1` performs the external-band sum directly, avoiding large cancelling internal terms.

Figures 1 and 2 describe the same occupied band space. Their numerical values need not coincide at this resolution: the FHS method gives finite-plaquette averages, while this Kubo result uses pointwise canonical momentum and a finite empty-band window. The 12×12 mesh and 26 stored bands require convergence for quantitative predictions. NumPy interpolation smooths only the display; all integration and numerical checks use the original samples.

Calculation and interpretation

3.2.1 MoS₂: an isolated single band near the valleys

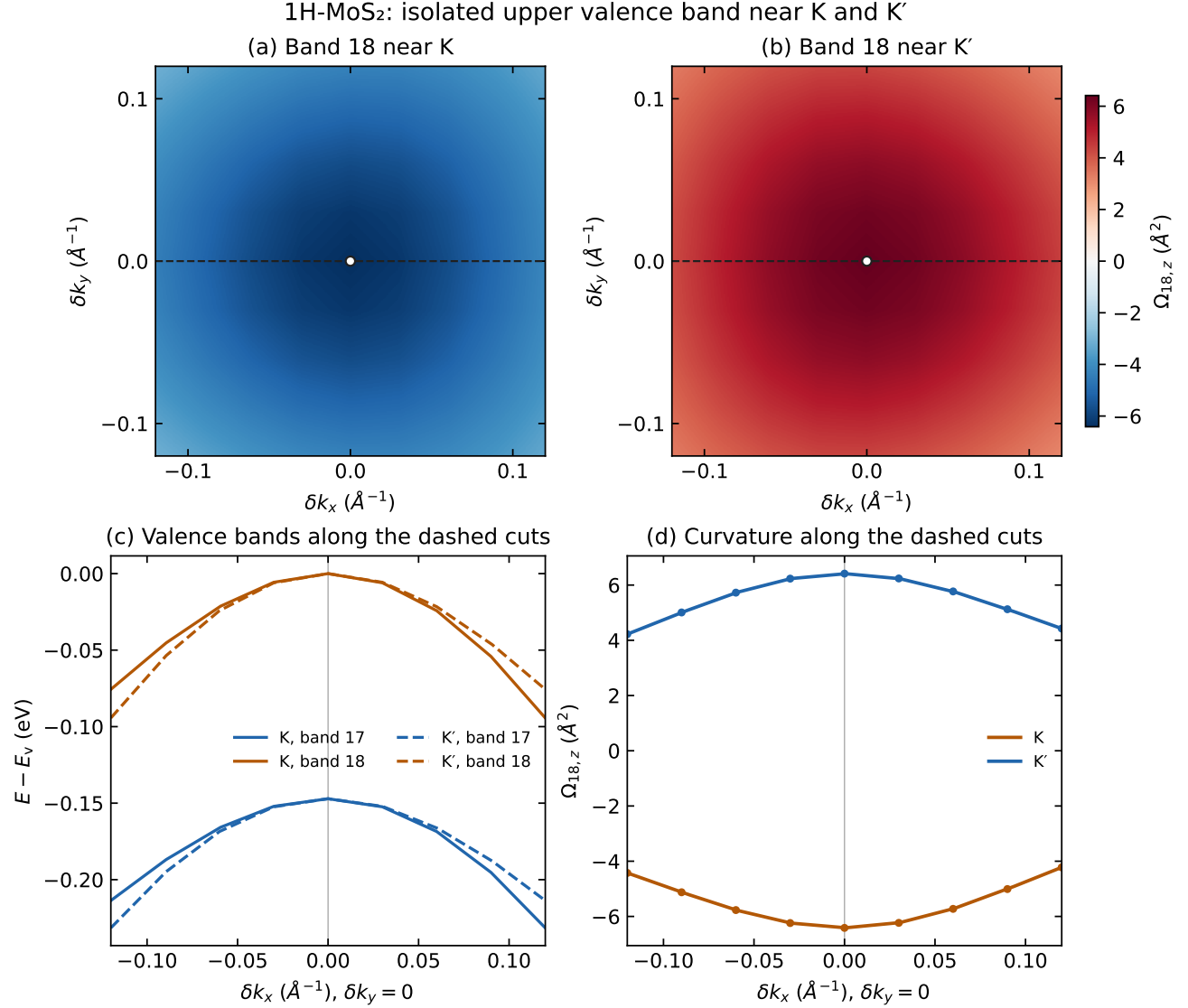


Figure 3. Single-band Kubo curvature of the upper valence band, band 18, around K and K'. Each local Cartesian patch extends $\pm 0.12 \text{ \AA}^{-1}$ and contains 9×9 VASP points. The upper maps use NumPy bilinear display interpolation; the dashed lines mark the cuts below. The lower panels show the two upper VASP valence bands near both valleys and band-18 curvature at both valleys. Symbols mark the calculated curvature samples.

Band 18 touches its Kramers partner at Γ , but is separated from every other stored band by at least **0.130 eV** throughout these valley patches. The 17–18 splitting at K is **0.147 eV**, and the valley curvatures are approximately **$-6.415/+6.415 \text{ \AA}^2$** at K/K'. This is a suitable local single-band example of the spin-split valley physics described by Xiao et al.. The patch data describe local curvature; a full-zone integral requires the appropriate isolated band bundle and complete BZ sampling.

VASP preparation, calculation and reference data

3.2.2 MoS₂: intrinsic charge and regional valley Hall response

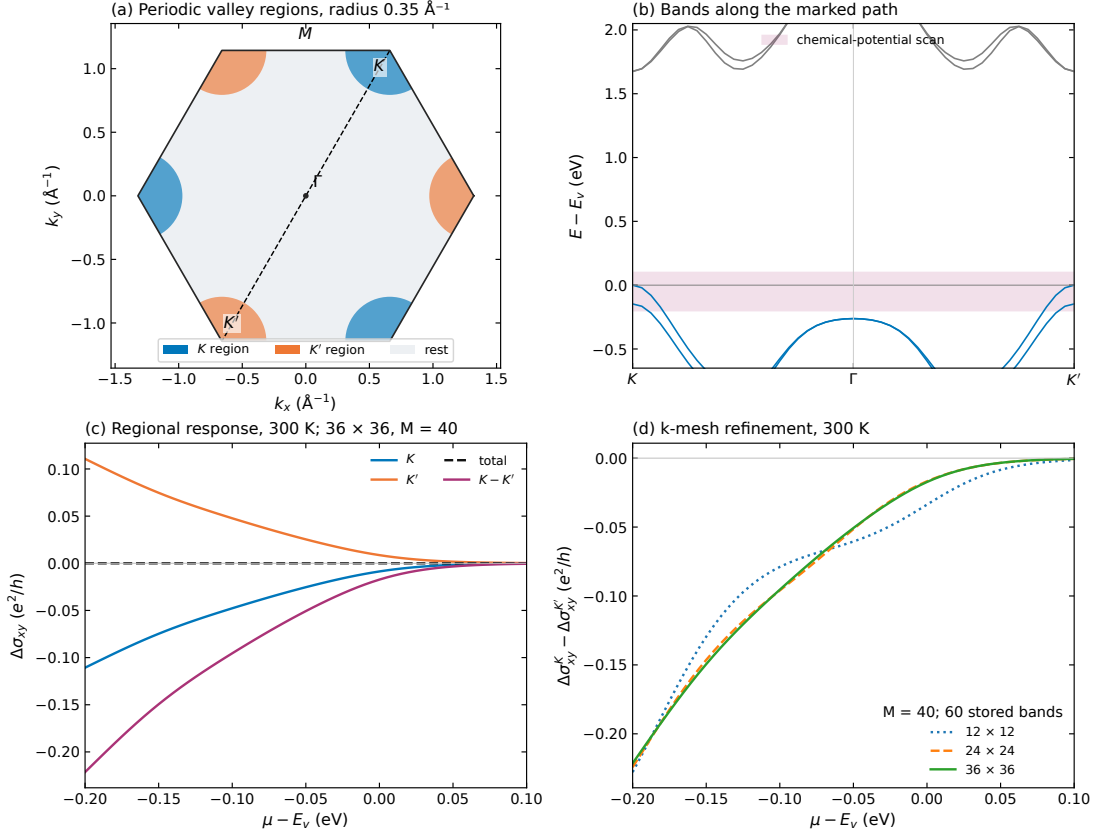


Figure 4. (a) Cartesian first BZ with periodic K/K' disks of radius $0.35 \text{ \AA}^{-1}$ and the marked K- Γ -K' line. (b) The matching 26-band VASP dispersion, with the chemical-potential interval shaded. (c) Regional Hall changes at 300 K on the 36×36 mesh, using 60 stored bands and pairs within bands 1–40. (d) Mesh refinement at that fixed band window. Energies are relative to each source's valence-band maximum; the band path uses the same density and cutoff.

The native Fortran calculation supplies the interband pairs; Python post-processing applies occupations and integrates their contribution over the full zone and the specified regions. The 61-point scan covers $E_v - 0.20$ to $E_v + 0.10$ eV at 0 and 300 K. We define $\Delta\sigma(\mu) = \sigma(\mu) - \sigma(\mu_{\text{ref}})$ with a midgap reference, and the valley difference as $\Delta\sigma_K - \Delta\sigma_{K'}$. The total charge response remains below $2.24 \times 10^{-9} e^2/h$, consistent with time-reversal symmetry. Numerical tables also give absolute conductivities and carrier counts.

Two numerical choices were varied separately at 300 K. The relative L2 difference below is the norm of the curve change divided by the finer curve's norm, on a common $\mu - E_v$ grid.

Test	Refinement	Relative L2 change
k mesh; 60 stored bands, pair cutoff 40	$12 \times 12 \rightarrow 24 \times 24$	12.89%
k mesh; same band window	$24 \times 24 \rightarrow 36 \times 36$	1.482%
Pair cutoff; one 12×12 , 96-band source	$60 \rightarrow 80$	1.263%
Pair cutoff; same source	$80 \rightarrow 90$	0.202%

The final mesh change exceeds the stated 1% criterion. The last cutoff increment is below 1%; these separate tests do not establish joint convergence. The reference guide includes the commands and additional checks on empty-state accuracy and numerical degeneracy grouping.

This rigid-band response uses canonical momentum, omitting PAW augmentation and nonlocal/SOC velocity terms. Numerical refinement does not remove those approximations. The supplementary plot retains the finite-mesh steps at 0 K.

3.2.3 Optional comparison: matching the current operators

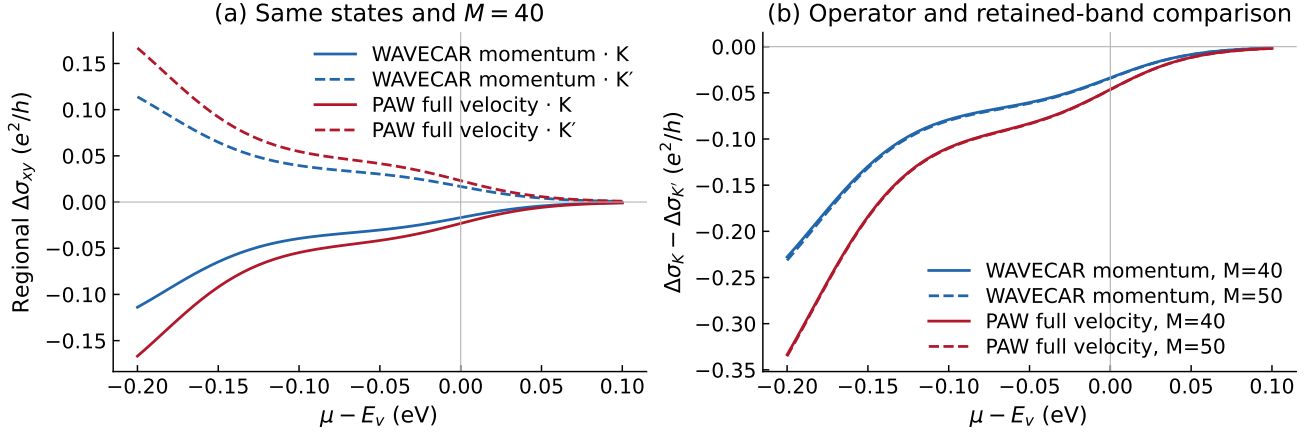


Figure 5. Charge response from the same final VASP eigenstates on a 12×12 mesh, with 60 stored bands, at 300 K. (a) Regional changes for K and K' using pairs within bands 1–40. (b) The K–K' difference with retained cutoffs $M = 40$ and 50. Blue curves use ordinary WAVECAR canonical momentum; red curves use the optional full PAW velocity matrices. Geometry, density, occupations, region definitions and eigenstates are identical.

At $\mu - E_v = -0.20$ eV and $M = 40$, the regional difference changes from -0.22779 to -0.33363 e^2/h when the full velocity replaces canonical momentum. Both total charge responses remain below 1.1×10^{-8} e^2/h , without time-reversal averaging. The nonzero regional response therefore reveals an operator difference that the symmetry-enforced total cannot test.

For the complete 300 K difference curve, increasing M from 40 to 50 changes the canonical and PAW results by **1.445%** and **0.465%**, respectively, using the same relative-L2 definition as above. At $M = 50$, the two operators differ by 29.44% relative to the PAW curve. These measurements describe one fixed mesh and band increment. They do not establish k convergence or a general advantage in convergence rate.

The full-velocity route includes the supported PAW, nonlocal and SOC operator terms, giving a more complete current description. Generating its input uses the optional VASP instrumentation; ordinary WAVECAR remains the starting workflow. The separate comparison guide provides preparation instructions, both reusable pair caches and the original CSV/DAT/NPZ results. Their integration can be repeated without VASP.

3.3 MoS₂: circular optical selection across layer stackings

The native optical calculation resolves left- and right-circular transitions from the WAVECAR momentum matrix elements. Circular selectivity is their intensity difference divided by their sum. The same analysis can be repeated with standard VASP optical matrices as an optional comparison.

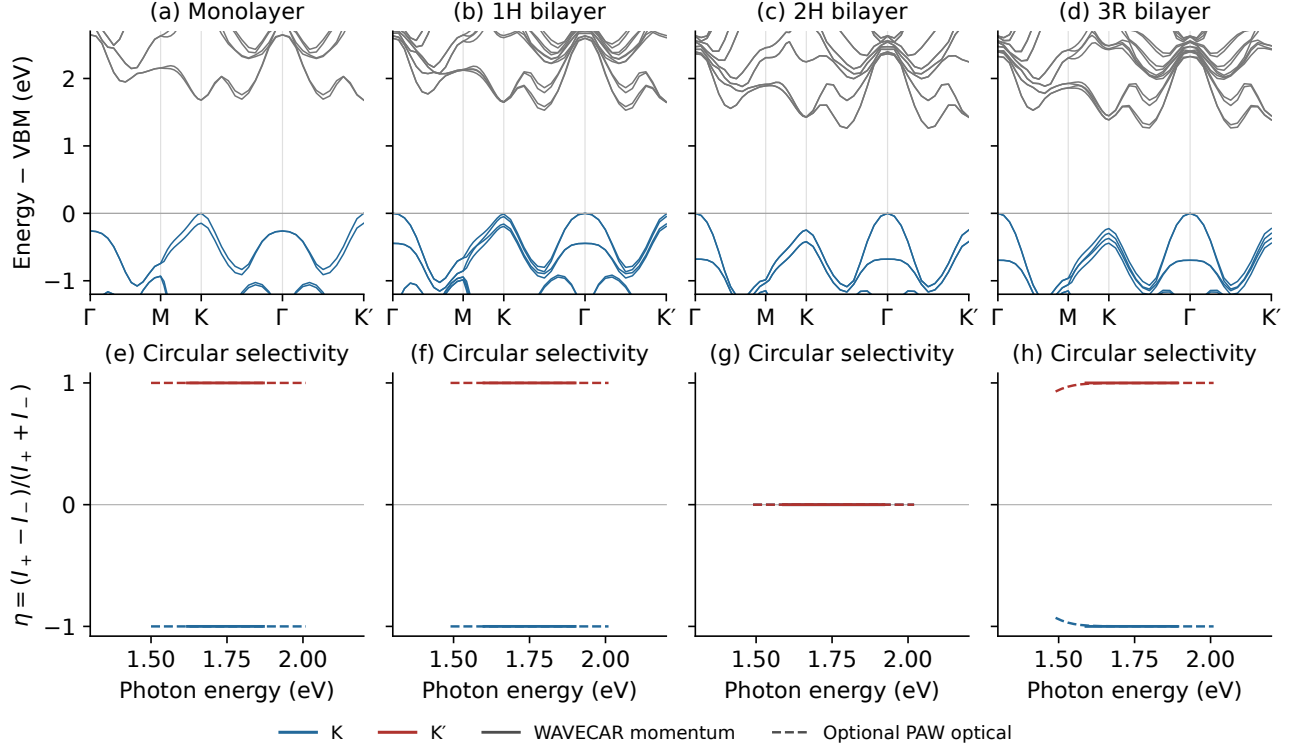


Figure 6. (a–d) VASP bands of monolayer and 1H, 2H and 3R bilayer MoS₂ along Γ –M–K– Γ –K', with each case referenced to its own sampled VBM. (e–h) K/K' circular selectivity under normal incidence, from ordinary WAVECAR momentum (solid) and optional standard WAVEDER matrices (dashed). Transitions use occupied bands 1–18 to 19–20 for monolayer, and 1–36 to 37–40 for all bilayers, with Gaussian $\sigma = 0.05$ eV. Weak spectra and native ratios limited by printed intensity precision are masked; overlapping curves retain their calculated values.

The near-edge peaks lie at **1.67–1.68 eV**. Monolayer and 1H show $\eta \simeq -1/+1$ at K/K'; the 3R PAW peak gives **–0.99884/+0.99884**. For the inversion-symmetric 2H control, summing the complete occupied and empty groups cancels the selectivity: the PAW peak residual is below 5×10^{-7} at both valleys. This illustrates why the response of an unresolved degenerate group must include every partner.

The stacking comparison extends one optical workflow to several VASP input systems. It is motivated by the 1H bilayers reported by Yang et al., using fixed idealized geometries for the present software demonstration. It does not reproduce the paper's direct-gap prediction or measured photoluminescence polarization. Earlier figures already demonstrate curvature and Hall integration; those maps are not repeated for each stacking.

The optional PAW optical calculation uses ordinary VASP output and requires no source modification. Its comparison with native momentum tests helicity and symmetry. The two spectral definitions have different finite-broadening weights, so their ratio differences do not isolate an operator correction and their absolute intensities are not equated. The stacking guide and reference spectra provide complete VASP inputs, both channel spectra and reproduction commands; the original monolayer optical tutorial remains the introductory example.

3.4 MoS₂ and Bi: contrasting native Z_2 invariants

Monolayer 1H-MoS₂ and buckled Bi illustrate trivial and quantum spin Hall insulators using the same native method. MoS₂ reuses the curvature example’s 12×12 WAVECAR; Bi uses the fresh PBE+SOC input described above. The Bi phase was proposed by Murakami. For each occupied spinor bundle, the Fukui-Hatsugai (FH) n-field method uses a time-reversal-compatible gauge and the half-zone integer-field parity:

$$\nu = \left[\sum_{p \in B_{1/2}} n(p) \right] \bmod 2.$$

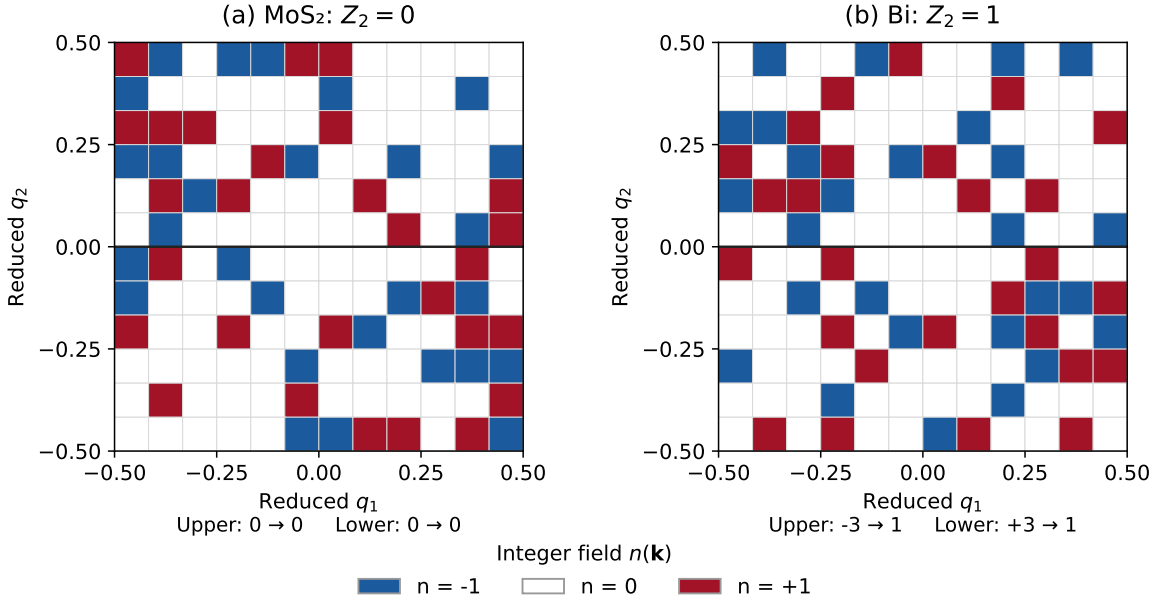


Figure 7. Native 12×12 n-fields for (a) monolayer 1H-MoS₂ and (b) buckled Bi. Red, white and blue denote +1, 0 and −1. Tiles are original plaquettes, without interpolation, in reduced coordinates $\mathbf{k} = q_1 \mathbf{b}_1 + q_2 \mathbf{b}_2$. Opposite edges are periodic; $q_2 = 0$ separates the half zones. Annotations show each sum followed by its parity.

Material	Occupied bands	Upper / lower sums	Z_2	Direct / global gap (eV)
MoS ₂	1–18	0 / 0	0	1.674 / 1.674
Bi	1–10	−3 / +3	1	0.535 / 0.497

Both occupied bundles have $\mathbf{C} = \mathbf{0}$ (Section 3.1 and the Bi check), yet their Z_2 values differ. Finite local Berry curvature and valley response are compatible with trivial MoS₂. The n-field pattern is gauge and branch dependent; its agreed half-zone parity is the invariant, not the local tile pattern or a Berry-curvature observable.

MoS₂ retains $Z_2 = 0$ at 24×24 with 60 stored bands, versus 26 at 12×12; both select occupied bands 1–18. Serial and MPI field CSVs are byte-identical at each mesh. This checks invariant stability under both input changes, without isolating mesh convergence or establishing transport convergence.

The native PASS checks use WAVECAR pseudo-wavefunction overlaps. Independent raw occupied-subspace time-reversal checks are retained with the references; native reconstruction alone does not establish physical symmetry. Optional Bi edge connectivity supports its bulk result (Appendix B.2). Conventional spin Hall response is a separate observable and need not be quantized when spin is not conserved (Appendix A.3).

Commands and paired figure · MoS₂ native field · Bi native field · Fresh Bi preparation

3.5 MoS₂: a real-space Γ state

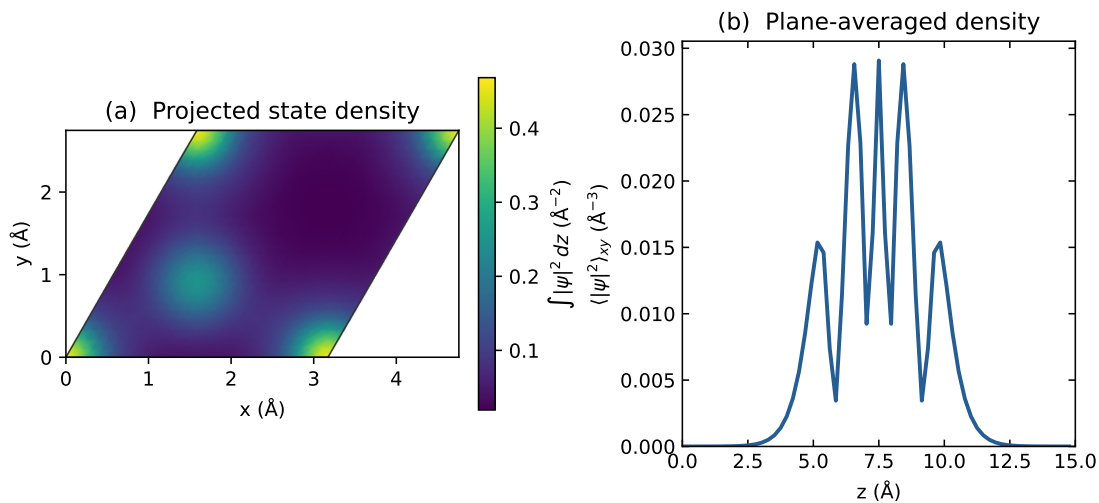


Figure 8. Projected pseudo-wavefunction density of band 18 at Γ and its plane average along z . The in-plane plot uses the Cartesian geometry of the oblique primitive cell. Both spinor components are retained.

The integrated pseudo-density is approximately **0.8258**, consistent with the plane-wave coefficient norm. The PAW augmentation is not part of this plot, so a unit density integral is not imposed. The example illustrates conversion of complex amplitudes into a physical-coordinate density map.

Wavefunction calculation guide

3.6 MnBi₂Te₄: occupied-space Chern number from the FHS method

This example demonstrates a nonzero Chern number of the complete occupied VASP band space. The three-septuple-layer MnBi₂Te₄ film has 21 atoms with +z/-z/+z Mn magnetization. Its fixed, unrelaxed geometry retains bulk vertical spacings from Yan et al., with in-plane $a = 4.336 \text{ \AA}$, PBE+SOC and Mn $U_{\text{eff}} = 5.34 \text{ eV}$ following Otrokov et al. It is distinct from the relaxed films in that study.

The VASP gap is **17.10 meV**. The FHS method applied to the **complete occupied valence-band space, bands 1–123, gives the Chern number $C = -1$** , corresponding to $\sigma_{xy} = +e^2/h$ in our convention. A direct 6×6 PAW optical integral instead gives **163.109 e^2/h** : the Γ -point curvature reaches approximately $-1.52 \times 10^4 \text{ \AA}^2$, and this coarse mesh overweights the narrow peak. This optical result is a convergence diagnostic, not a quantized response. An integer Chern number obtained with the FHS method does not establish convergence of the pointwise Kubo integral, and no integer value is imposed on that integral.

The native Fortran calculation evaluates overlap determinants directly in the complete VASP occupied space. Its $C = -1$ result and all 36 plaquette fluxes agree with the independent Python implementation of the FHS method within the native map’s printed precision. The separately checked deep bands 1–36 form a $C = 0$ bundle. The native output and comparison and material guide provide the actual command, VASP inputs, overlap diagnostics and standard-WAVEDER coarse-mesh result.

The main lesson is the distinction between occupied-bundle topology and response integration. The optional dense Wannier calculation in Appendix B.3 checks the expected Hall value within a supplied finite model; it is not needed to perform the direct-wavefunction workflows above and does not establish convergence of this coarse VASP optical integral.

3.7 Graphene: Chern numbers of projected-spin sectors

The graphene example uses actual VASP spinor wavefunctions for a planar two-atom cell with $a = 2.46 \text{ \AA}$ and a $20 \text{ \AA}$ cell height, PBE+SOC, a 520 eV cutoff and 16 stored bands. The selected space comprises bands 1–8. With Cartesian +z as the analysis axis, the native 6×6 , 9×9 and 12×12 calculations give $C_+ = +1$, $C_- = -1$, $C_{\text{charge}} = 0$ **and** $C_{\text{spin}} = 1$, within floating-point precision. Each sector has rank four; the minimum absolute projected-Pauli eigenvalue is 0.99999947.

Full mesh	Minimum direct gap (μeV)	C_+	C_-	C_{spin}
6×6	0.9120148	+1	-1	1
9×9	0.9119943	+1	-1	1
12×12	0.9120028	+1	-1	1

The sampled direct gap is only **0.9120 μeV** . A separate K/K' no-SOC control gives splittings below 0.00023 μeV . Tightening the fixed-density eigensolver's EDIFF from 1e-9 to 1e-10 eV changes the 6×6 minimum gap by approximately 5.4e-12 eV. These checks distinguish the resolved gap from that numerical floor; they do not establish convergence with respect to every DFT setting. The full no-SOC 6×6 input is rejected by the native energy-isolation check: its minimum gap is 1.94×10^{-10} eV, below the default 10^{-8} eV threshold, and no CSV containing spin Chern numbers is produced.

Reproducing this tiny gap requires the exact generating geometry in the CHGCAR header. The preparation script restores that header from the matching POSCAR while preserving the density and augmentation payload byte for byte; the example records both hashes. Rounded coordinates in the fixed-density input can otherwise introduce a spurious splitting. VASP's finite smearing gives fractional occupations near K. The Chern-number calculation uses the gapped eight-band geometric projector; the source occupations are reported separately.

The Chern numbers use the WAVECAR pseudo metric and projected-spin links of Section 1.1.1. They are not a conventional spin Hall conductivity. The example supplies VASP preparation, native CSVs and optional plotting commands; the CSVs are also directly usable in other plotting tools. GNU serial/MPI CSVs agree byte for byte on all three meshes. Intel and GNU sector fluxes agree with the independent NumPy evaluation within 1.5×10^{-14} rad.

Graphene: Chern numbers of projected-spin sectors

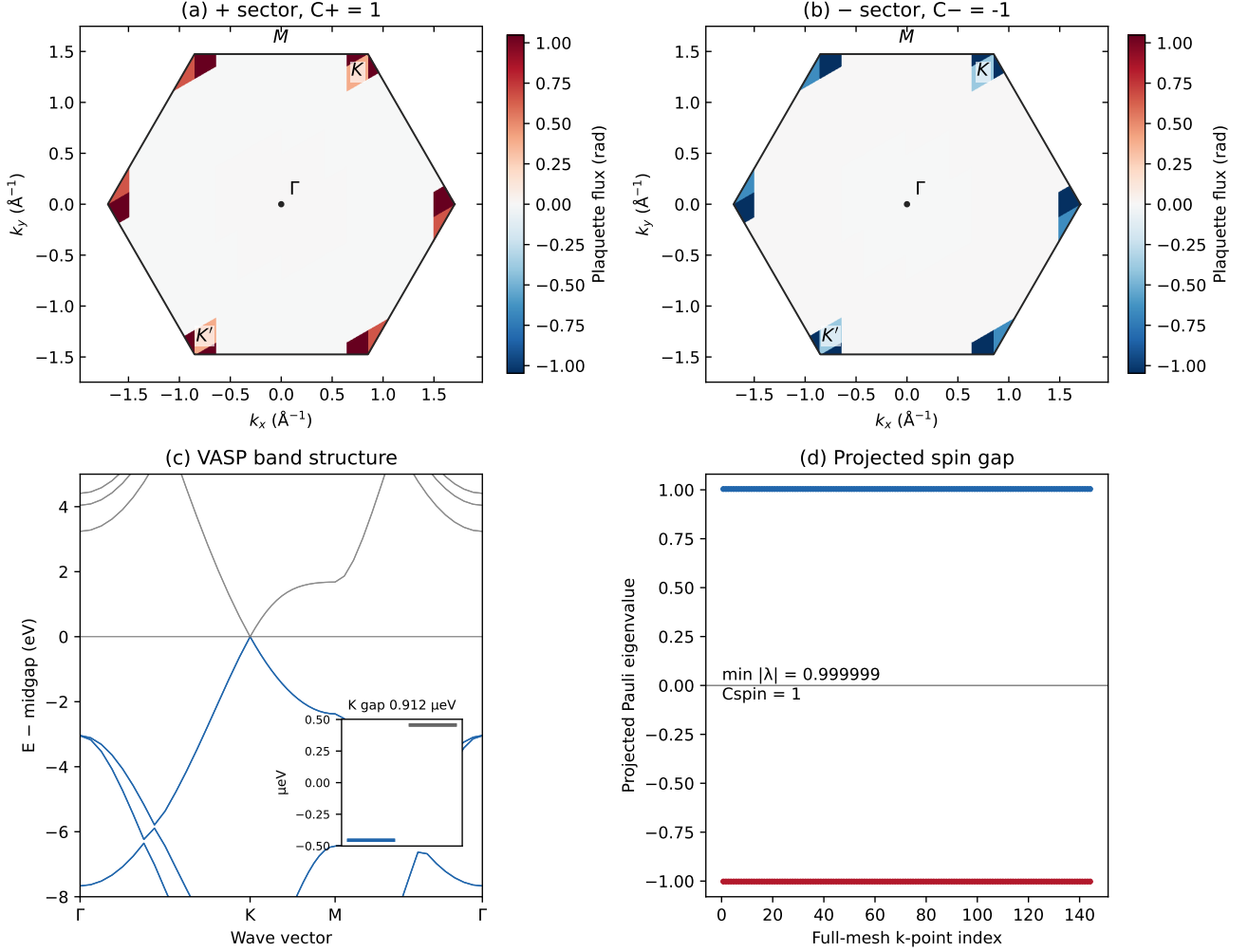


Figure 9. (a,b) Native positive- and negative-spin-sector plaquette fluxes on the 12×12 mesh, shown without interpolation. The sectors carry opposite Chern numbers while the charge sum vanishes. (c) Actual VASP bands on Γ -K-M- Γ ; the inset resolves the 0.912 μeV K-point gap using full-precision WAVECAR energies. (d) The projected-Pauli spectrum remains separated from zero. Mesh agreement establishes the integer result for this fixed input; the maps do not resolve the microscopic width or peak height of the Dirac curvature. Saved data and drawing commands reproduce all four panels.

3.8 Graphene: spin-sector Kubo curves and a tiny-gap limit

The spin-Kubo example uses the same graphene Hamiltonian as Section 3.7, with a 49-point Γ -K-M- Γ path and a separate 12×12 mesh. Execute VASPBERRY in each directory containing its matching WAVECAR and OUTCAR:

```
# In the path directory: local curvature, no Chern number assigned.
mpiexec -n 4 "$VB" --task spin-kubo --bands 1:8
# In the separate full 12x12 mesh directory:
mpiexec -n 4 "$VB" --task spin-kubo --bands 1:8 --mesh 12,12
mpiexec -n 4 "$VB" --task spin-chern --bands 1:8 --mesh 12,12
```

Here VB is the native executable built as described in the Intel MPI walkthrough. OUTCAR supplies the actual source spin frame; spinor coefficients and energies come from WAVECAR. Neither a PROCAR spin weight nor a manual spinor-count option is used.

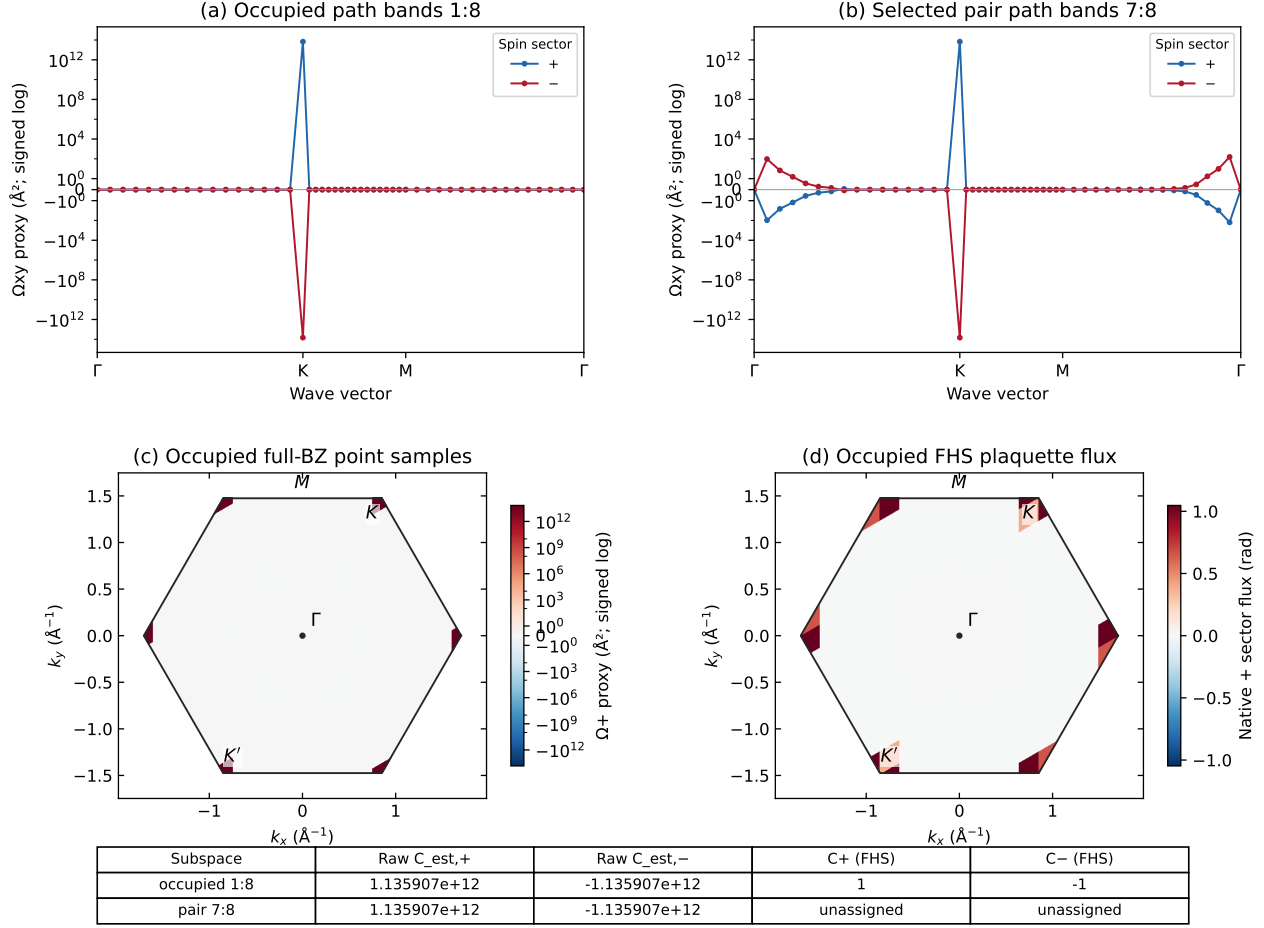
For the complete occupied 1–8 space, the geometric result remains $C_+ = 1, C_- = -1$. However, the raw 12×12 point-curvature estimate gives $C_{\text{spin}}^{\text{est}} = 1.135906684114 \times 10^{12}$. It is **not a Chern number**. The roughly micro-eV gap creates an extremely narrow structure, and the present sampling does not resolve its integral. Finite source bands and the canonical/pseudo operator approximation also remain. These data do not isolate the contribution of each error. Even the charge cancellation has an absolute residual near 1.44×10^4 against sector magnitudes near 10^{12} ; it is not evidence for a large physical charge response.

The tiny gap also conditions numerical reproducibility. GNU serial and MPI path tables are identical in the checked run. Intel Classic reproduces the dominant K-point sector peaks to about 4×10^{-15} relative accuracy, but cancellation-sensitive parent and transverse entries do not meet a uniform 10^{-9} relative tolerance. Such agreement of the dominant peaks does not imply accuracy of their much smaller difference.

Selecting the top pair with `--bands 7:8` produces local sector curvature where the sampled energy and spin gaps pass. A separate full-mesh FHS calculation rejects that pair: an adjacent-sector overlap has minimum singular value 2.35×10^{-10} , below the link threshold. Consequently the example retains the rejection and assigns **no pair Chern number**. Pointwise gap checks alone do not certify globally resolved connectivity; the exact cause of this pair-link failure is not established here.

Graphene: occupied and selected-pair spin sectors

Canonical-momentum derivative proxy · pseudo Gram metric · no integer rounding



Lines connect stored points; cell colors are constant. C_est is not a certified topological invariant.

Figure 10. (a) Positive/negative occupied-sector Kubo curvature along Γ -K-M- Γ . (b) Local curvature of the 7-8 pair, without a global Chern number assignment. (c) Raw occupied positive-sector point curvature on the 12×12 mesh. (d) Independent positive-sector Berry flux from the FHS method. Signed-log display exposes both signs and the very large peaks; it does not modify the stored data or repair the mesh integral. The example plotter reads native CSVs, changes no wavefunctions and performs no additional curvature calculation. Reference data and input/output hashes accompany the figure.

3.9 Bi: occupied sectors and an isolated top pair

The Bi spin-sector example uses the fresh fixed density of Section 3.4, 48 stored bands, a 400 eV cutoff, a complete 6×6 mesh and a separately calculated 49-point Γ –K–M– Γ path. For this positive- 60° lattice convention, fractional K is $(1/3, 2/3, 0)$ and M is $(0, 1/2, 0)$. The guide supplies ordinary VASP path preparation and the exact native execution commands. At shared path/mesh points, occupied energies agree within 5.2×10^{-11} eV. The highest stored empty band differs by up to 4.73×10^{-4} eV; total-energy convergence alone does not establish convergence of every intermediate state in the Kubo sum.

The occupied bands 1–10 split into rank-five sectors with sampled $C_+ = 1, C_- = -1$, consistent with the independently evaluated 2D Bi-bilayer $Z_2=1$. The isolated top pair 9–10 instead splits into rank-one sectors with sampled $C_+ = 2, C_- = -2$. These are different projectors: a pair invariant does not replace the full occupied-system Z_2 . For the pair, the minimum sampled external energy gap is 0.0474281 eV and the projected-spin gap is 0.3348024. Its maximum plaquette flux is 2.89067 rad and minimum sector-link singular value is 0.04317. These diagnostics belong to the initial 6×6 reference. Section 3.10 follows the same Hamiltonian through 12×12 and 18×18 meshes; it separates the stable discrete invariant from the slowly resolved Kubo quadrature.

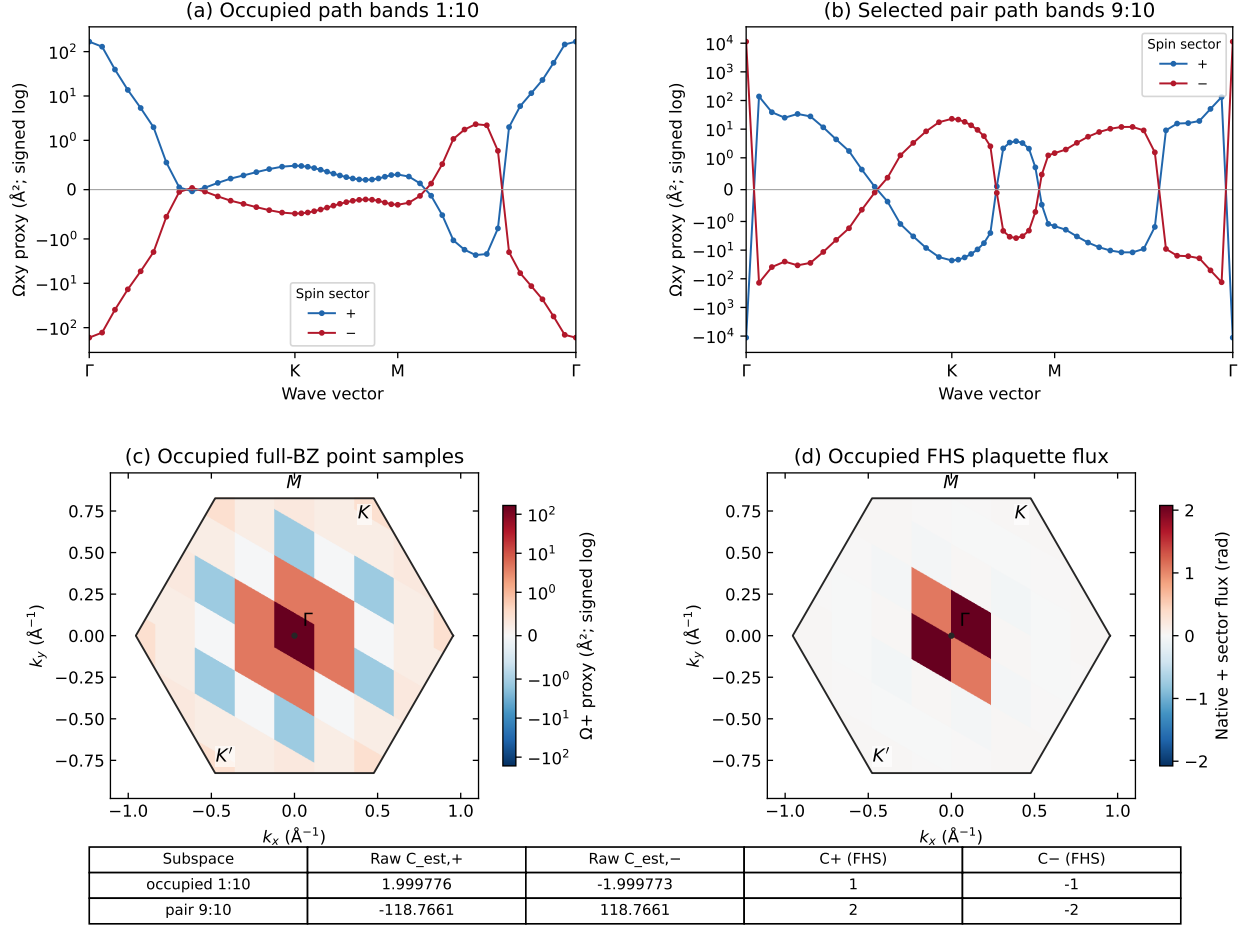
Bi subspace	Sector ranks	Sector Chern numbers	Raw $C_{\text{spin}}^{\text{est}}$
		C_+, C_- (FHS)	
Occupied bands 1–10	5 + 5	+1, −1	1.999774784
Top pair 9–10	1 + 1	+2, −2	−118.7661327

The Kubo estimates disagree strongly with the geometric results. Rounding 1.99977 to 2 would even assign the wrong occupied-sector invariant. Keep the Chern number from the FHS method and the local canonical approximation separate; refine sampling and source bands, then examine the missing physical-operator terms before using the Kubo integral quantitatively.

Use `--bands 1:10` for the occupied result or `--bands 9:10` for the pair, on both the path and the full mesh. The spin axis must also match. Positive and negative labels refer to the projected-Pauli eigenvalue; for generic SOC they do not identify two independent collinear energy channels.

Bi: occupied and selected-pair spin sectors

Canonical-momentum derivative proxy · pseudo Gram metric · no integer rounding



Lines connect stored points; cell colors are constant. C_est is not a certified topological invariant.

Figure 11. (a,b) Actual 49-point occupied and pair-sector path curves. (c) Occupied positive-sector point curvature on the 6×6 mesh. (d) The same occupied space's independent positive-sector Berry flux from the FHS method. All panels use native outputs. Straight segments connect path samples only; no interpolated Hamiltonian supplies extra samples. The saved decomposition columns permit additional plots of the parent contribution and the projected-spin mixing correction.

3.10 Follow-up convergence tests

3.10.1 Bi: separate mesh, response cutoff and source-state quality

The convergence example holds the structure, PAW potential, fixed SCF density and physical Hamiltonian constant. Three controls answer different questions: vary the mesh with source NBANDS=64 and --sum-bands 48; vary the sum cutoff on one unchanged 80-band WAVECAR; and compare source64/source80 at the same retained cutoff. The example supplies ordinary VASP preparation, native MPI commands and portable full-point tables, with separate source and output check-sums.

Mesh, source64 / cutoff48	Occupied raw spin integral	Pair raw spin integral	Spin Chern number (FHS), occupied / pair
6×6	1.9997800132	-118.76612842	1 / 2
12×12	1.0382696725	-29.02711625	1 / 2
18×18	1.0268699256	-12.25375432	1 / 2

The occupied proxy moves toward the discrete value but still changes by 0.01140 between the last two meshes and differs from 1 by 0.02687. The pair proxy is plainly unconverged. Its Γ sample alone contributes -118.5620, -29.6405 and -13.1736 under the shrinking uniform-cell weights. These are single-point quadrature contributions, not integrals over a resolved Γ region. The actual pair path changes from about -11331.8 $\text{\AA}^2$ at Γ to +138.5 $\text{\AA}^2$ only 0.05965 $\text{\AA}^{-1}$ away; that path does not resolve a 2D peak.

The independent full-mesh overlap calculation gives occupied $C_+ = 1, C_- = -1$ and pair $C_+ = 2, C_- = -2$ on all three meshes. At 18×18 the pair's minimum link singular value rises to 0.35449 and its maximum plaquette flux falls to 0.94453 rad, compared with Section 3.9. This supports stable sampled sector topology; it does not make the pair proxy integral quantized or turn its invariant into the occupied Z_2 .

At fixed 6×6 and cutoff48, source64→source80 changes the occupied integral by 1.6×10^{-8} and the pair by 1.1×10^{-6} . Common-point curvatures agree within 2.2×10^{-8} of the shared peak across the source64 meshes. The first 48 eigenvalues in the source64 and source80 controls agree within 6.5×10^{-11} eV, while bands 49–64 can differ by about 0.12 eV. Total-energy stopping criteria do not certify those high empty states. Increasing the sum to include them mixes response truncation with a source-state error.

The same-source80 cutoff sweep changes the occupied result by about 6.4×10^{-4} and the pair by -0.0040 from cutoff32 to cutoff48. That is a band-window check at fixed coarse sampling; it cannot repair the much larger pair mesh error. Extending the window requires adequately converged empty eigenstates as well as enough stored bands.

The archived sources partitioned each complete mesh. Native spin-kubo processed each partition with its original matching OUTCAR; post-processing validated the assembled periodic grid and applied the signed area weights of Section 1.2.1. The table is therefore labelled as an aggregate of native point curvature, separately from native full-mesh integral output. The independent validation with the FHS method uses actual assembled WAVECAR coefficients and the common verified spin frame. Source headers were not altered.

An additional ordinary VASP 12×12/source64 restart produced a genuine matching full-mesh WAVECAR/OUTCAR pair. Native spin-chern independently recovered occupied-space spin Chern number 1 and selected-pair spin Chern number 2. Native cutoff48 integrals differ from the partition aggregate by only 8.19×10^{-9} (occupied) and 2.98×10^{-7} (pair). This checks the assembly and quadrature route; it does not certify the physical convergence of those values.

Bi: sampling and source-band controls remain distinct

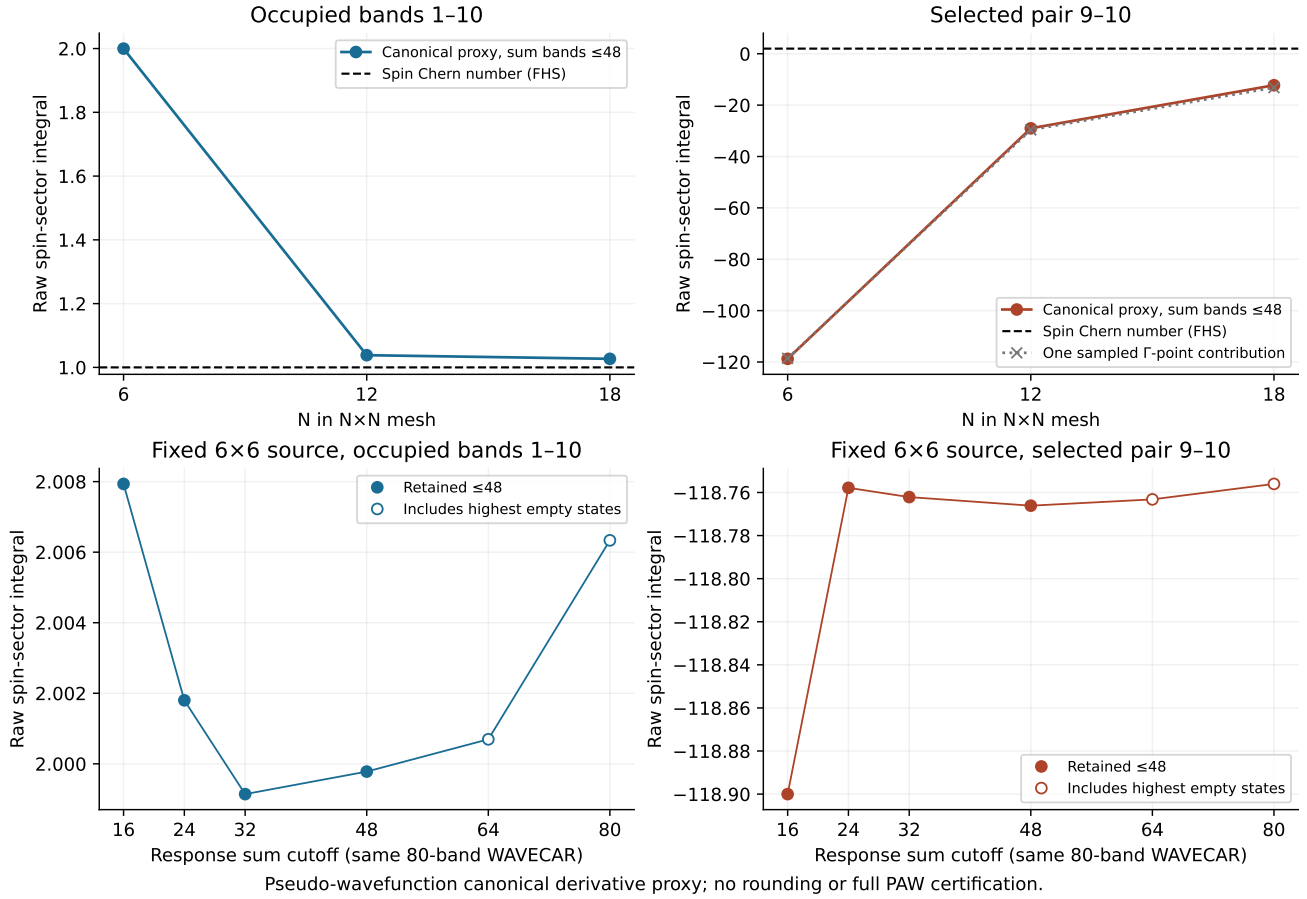


Figure 12. Bi mesh and intermediate-state controls using the same Hamiltonian. The figure retains noninteger and negative raw estimates; Spin Chern numbers evaluated with the FHS method are shown as independent geometric references. The cutoff sweep uses an unchanged source80 WAVECAR. The associated source-padding table holds the retained cutoff fixed. Full point data permit alternative plots without rerunning the native curvature calculation.

3.10.2 Graphene: actual local sampling around the SOC gap

The local graphene example adds 146 actual VASP points around K and K' to the same fixed-density, 520 eV, 16-band calculation. Each valley has its center and nine radii from 10^{-9} to 10^{-5} Å⁻¹, with eight angles per ring. `spin-kubo --bands 1:8` writes the local Cartesian curvature directly; no mesh option is used because these points do not form a full BZ mesh. An independent diagnostic reads the same WAVECAR and constructs small projected-spin polygon loops, retaining the full selected Gram matrix.

The measured center gap is about 0.9120 µeV. Fitting only the measured energy dispersion to $E(q)^2 = m^2 + v^2 q^2$, where $E(q)$ is half the band 9–8 separation and m its value at the valley center, gives $v \simeq 5.479$ eV Å and a characteristic radius $q_0 = m/v \simeq 8.323 \times 10^{-8}$ Å⁻¹. This is the scale that a local curvature quadrature must resolve. The dashed massive-Dirac curves in Figure 13 are diagnostics fitted to these energies. Their positive sign is chosen to match the measured sector; energies alone do not determine chirality. No curvature amplitude or target integer Chern number is fitted.

At radius 10^{-5} Å⁻¹, the actual eight-vertex polygon spin flux divided by 2π is about 0.49561 in each valley, with a two-valley sum of 0.99122. The polygons occupy only local regions and are not the full BZ. Four- versus eight-vertex polygons change the enclosed region and its straight edges; agreement cannot by itself certify circular integration. Every sampled fan-triangle phase stays below the explicit $\pi/2$ guard, and the minimum overlap singular value is 0.70713. These checks avoid an unresolved sampled branch crossing without certifying unsampled spatial structure.

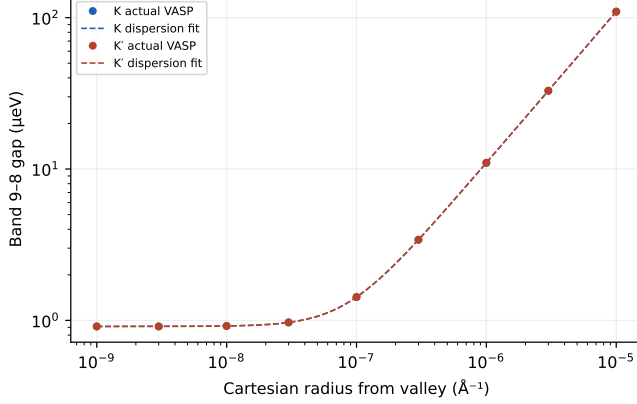
Native curvature at the valley center is about 6.8216×10^{13} Å². An independent small-polygon flux divided by its actual polygon area at radius 10^{-9} Å⁻¹ gives approximately 7.2175×10^{13} Å². The native value is about 5.5% smaller. It is also 0.94507 times the energy-fit Dirac center curvature. Reducing the native intermediate-state cutoff from 16 to 10 on the unchanged WAVECAR changes the sector curvature by at most 1.13×10^{-10} relative to the sector magnitude. This particular cutoff check does not explain the residual difference, nor does it exclude contributions from states not present in the 16-band source or missing physical-operator terms.

A stricter 50-point repeat at the same coordinates changes EDIFF from 10^{-9} to 10^{-11} eV and NELMIN from 20 to 40; it reaches the stated stopping criterion after 47 Davidson steps. The maximum gap change is 7.56×10^{-11} eV and the maximum relative native spin-curvature change is 1.56×10^{-4} (0.0156%). Local geometric spin flux divided by 2π changes by at most 1.52×10^{-9} . This solver sensitivity is much smaller than the approximately 5.5% curvature difference; it does not test the fixed SCF density or PAW model.

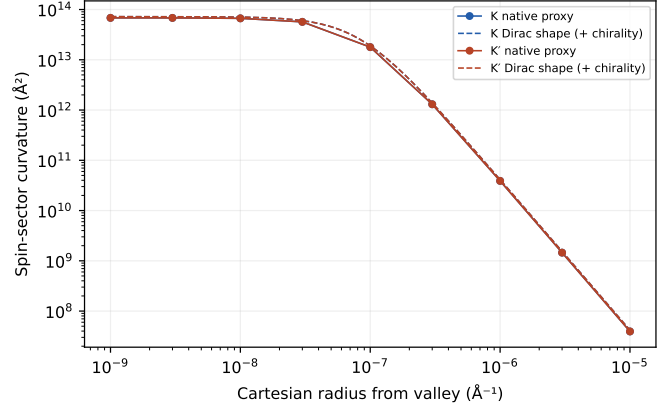
The Python diagnostic integrates the native point table using the angular average and a trapezoid rule in log radius. This is distinct from the native full-mesh integral output. Keeping five versus nine radial nodes changes the per-valley result from about 0.52084 to 0.47021. These raw values demonstrate remaining radial quadrature error; neither is a converged Chern number. The energy-fit model's local disk result, actual polygon flux and native point quadrature are saved in distinct columns and labelled separately. The large uniform-grid estimate in Section 3.8 is therefore principally a sampling warning, while even resolved point values retain the declared canonical/pseudo derivative approximation.

A full-BZ adaptive calculation would need finer actual points near the peak width, disjoint valley and outer-BZ regions, correct area weights, and separate checks of radial, angular, electronic and retained-state convergence. Interpolating a plot or replacing an integral by the fitted Dirac value does not supply this calculation. For the occupied topological assignment, use the complete-mesh `spin-chern` result of Section 3.7.

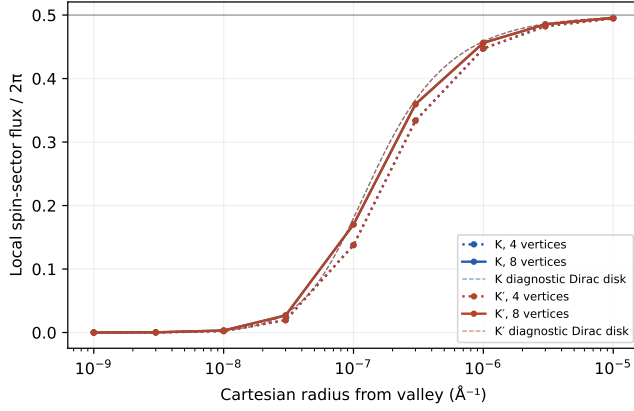
Graphene intrinsic SOC: local resolution study
 Fixed 520 eV / 16-band SCF density; neither local integration is a full-BZ Chern number
 (a) Actual microscopic gap scale



(b) Native curvature and diagnostic model



(c) Actual WAVECAR polygon flux / 2π



(d) Local native point quadrature

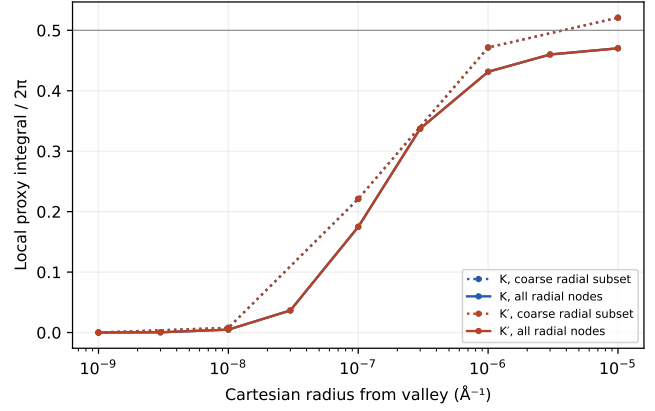


Figure 13. (a) Actual local VASP gap and an energy-only dispersion fit. (b) Native spin-sector curvature and the diagnostic Dirac shape. (c) Independent actual-WAVECAR polygon flux for different vertex counts, with the model's circular-disk flux labelled separately. (d) Local native point quadrature using coarse and full radial subsets. No panel presents a local or fitted quantity as a full-BZ invariant. Portable tables and ordinary VASP/native commands accompany the example.

4. Practical use

Begin with the example guide, reproduce the chosen quantity, and then select the sampling and band window appropriate to your own material. Use Cartesian reciprocal-space maps to display lattice symmetry, named k paths for band-resolved quantities, and energy or chemical potential for spectra and transport. State units, represented bands, temperature and broadening in the caption.

Start with WAVECAR and the native executable to calculate Chern numbers with the FHS method, the Z_2 invariant with the FH method, Kubo curvature, optical selection and state densities. For charge Kubo transport, export the native inter-band pairs and use the Python occupation/integration step to scan chemical potential and temperature without repeating the matrix-element calculation. These main workflows do not require Wannier90. For layer, orbital and spin attribution, add the matching PROCAR and OUTCAR, define atom groups and a spin axis, and follow the projection tutorial. Each strained, field-biased or magnetically distinct calculation needs its own matching wavefunctions and projections; changing occupations alone does not recalculate the electronic structure under those perturbations. Optional standard optical or full transition-matrix inputs allow additional operator terms to be assessed on matching electronic states. The operator guide links the input requirements and separate instructions for each route.

For quantitative conclusions, converge the VASP electronic structure, k mesh and relevant band window. The examples show how the methods are used; the supplementary analytic references check formulas and conventions, while version 1.4.0 validation and the earlier validation record document the numerical implementation. Detailed execution records remain alongside the numerical files for reproducibility.

Appendix A. Optional physical-operator extensions

These additional routes address the operator content of response functions. They complement the native WAVECAR examples; they are not prerequisites for Chern-number calculations with the FHS method, the FH Z_2 invariant, or canonical-momentum Kubo-formula analysis. Standard WAVEDER needs no VASP source change. Full velocity and PAW spin inputs use the separately documented instrumented producer. Their Python backends and convergence conditions are explicit below; the matched charge comparison remains in Section 3.2.3 so readers can assess its effect on the native baseline.

A.1 Standard VASP optical matrices

For an insulating occupied bundle, the supported standard VASP longitudinal optical calculation supplies matrix elements $C_{cv,\alpha} = \langle u_c | \partial_{k_\alpha} u_v \rangle$ in Å. Here v and c label occupied and empty states. Their contribution to the trace is

$$\Omega_{xy}^{\text{occ}} = -2 \text{Im} \sum_{v,c} C_{cv,x}^* C_{cv,y}.$$

The energy denominator is already contained in these derivatives. The longitudinal PAW optical expression includes projector and augmentation terms, as described by Gajdoš et al.. This standard WAVEDER route needs no VASP source modification. The `waveder-hall` command checks the actual VASP output, occupied filling and global gap before integration. Its present scope is VASP 5.4.4, zero temperature and a fixed insulating bundle; the occupied–empty separation must exceed the producer’s 2 meV degeneracy threshold. Mesh and empty-state convergence remain separate requirements.

The same standard optical matrices provide circular transition spectra at each $\mathbf{k}$ point. For propagation along $+z$ and $\epsilon_\pm = (\hat{x} \pm i\hat{y})/\sqrt{2}$,

$$I_\pm(\mathbf{k}, E_\gamma) = \sum_{v,c} \left| \frac{C_{cv,x} \pm iC_{cv,y}}{\sqrt{2}} \right|^2 g_\sigma(E_\gamma - E_c + E_v), \quad \eta = \frac{I_+ - I_-}{I_+ + I_-}.$$

Here g_σ is a normalized Gaussian. Complete initial and final band groups are summed before forming η , which is essential at degeneracies. These $\mathbf{k}$ -resolved strengths are independent-particle quantities; they do not include excitons or emission kinetics. The ordinary WAVECAR calculation uses canonical momentum with its native photon-energy normalization. The optical guide specifies both conventions and their comparison.

A.2 PAW spin matrices and conventional spin Hall response

The conventional spin current is $J_i^a = \{s_a, v_i\}/2$, with $s_a = \hbar\sigma_a/2$. In SOC systems its matrix elements require the full complex spin matrix, including off-diagonal band elements. Multiplying charge Berry curvature by a spin expectation value is generally insufficient. This current definition and its first-principles evaluation are discussed by Qiao et al. and Ryoo, Park and Souza.

The two WAVECAR spinor components directly give the pseudo matrix $\tilde{\Sigma}_{nm}^a = \sum_{Gss'} c_{nGs}^* (\sigma_a)_{ss'} c_{mGs'}$. The physical PAW matrix additionally requires the on-site correction

$$\Sigma_{nm}^a = \tilde{\Sigma}_{nm}^a + \sum_{Aijss'} p_{ni,s}^{A*} \Delta Q_{ij}^A (\sigma_a)_{ss'} p_{mj,s'}^A, \quad \Delta Q_{ij}^A = \langle \phi_i^A | \phi_j^A \rangle - \langle \tilde{\phi}_i^A | \tilde{\phi}_j^A \rangle.$$

Here p denotes the same-run PAW projector overlap. This expression is $T^\dagger \sigma_a T$ for the supported spin-independent PAW transformation. The implementation checks the corresponding physical overlap and preserves the raw pseudo-state norms. For this optional route, the supplied producer recipe instruments a separately licensed VASP 5.4.4 source copy. The ordinary WAVECAR and standard WAVEDER charge workflows do not need that instrumentation.

For transport, $D_i = \hbar v_i$ includes PAW and nonlocal/SOC velocity terms. The producer retains the velocity before the optical energy-denominator division, including diagonal and exactly degenerate blocks. These blocks cannot be recovered from an optical connection that has already discarded them. Within the supplied source-band space P , VASPBERRY forms $K_i^a = \{P\Sigma^a P, PD_i P\}/2$. This finite-band product omits the $P\Sigma^a QD_i P$ and $PD_i Q\Sigma^a P$ terms, where $Q = 1 - P$. Increasing the source-band window therefore tests both current-product closure and the intermediate-state sum.

For the complete occupied group $\mathcal{V}$, the implementation evaluates

$$\Omega_{ij}^a = -2 \operatorname{Im} \sum_{n \in \mathcal{V}, m \notin \mathcal{V}} \frac{K_{i,nm}^a D_{j,mn}}{(E_m - E_n)^2}, \quad \frac{\sigma_{ij}^a}{(\hbar/e)(e^2/h)} = \frac{A_{BZ}}{4\pi} \sum_{\mathbf{k}} w_{\mathbf{k}} \Omega_{ij}^a(\mathbf{k}).$$

Internal equal-occupation terms cancel before division. The formula uses $e > 0$, physical electron charge $-e$, Pauli matrices without an embedded $\hbar/2$, and D in eV Å. The curvature-like tensor is in Å². In a conserved $\sigma_z = +1$ sector, this normalized spin coefficient is minus one half of the charge coefficient in Section 1.2. No additional spin degeneracy or integer rounding is applied. The present command supports a fixed insulating 2D bundle at zero temperature. Its operator, gauge, units, sampled gap and band coverage are recorded with the numerical output.

Z_2 topology and conventional spin Hall response are separate tests. A nontrivial time-reversal invariant insulator need not have an exactly quantized bulk spin Hall coefficient when spin is not conserved. The optional ideal-edge check in Appendix B.2 addresses boundary connectivity, not a finite-device conductance. The spin Hall guide gives the operator contract and reproduction commands.

A.3 Bi: optional spin Hall conductivity and convergence

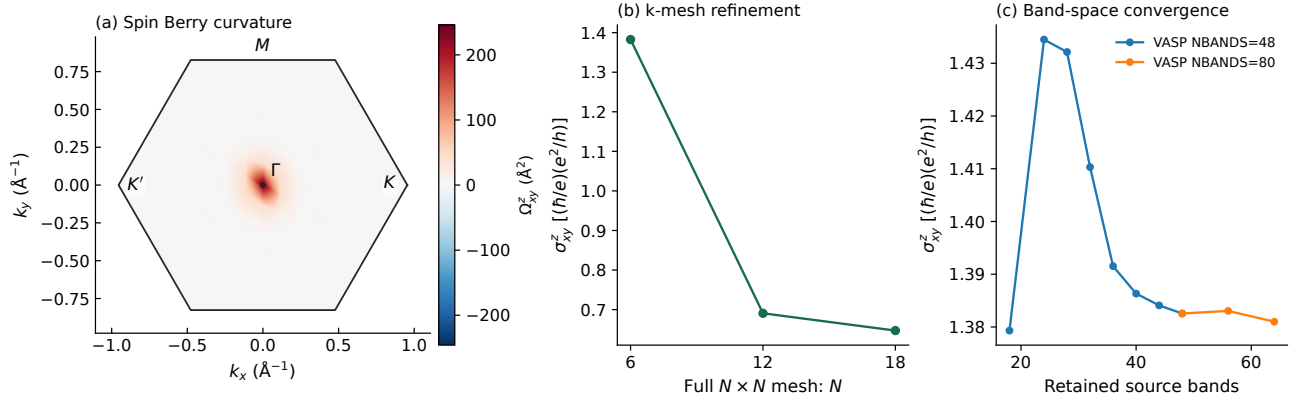


Figure 14. (a) Occupied-bundle spin Berry curvature Ω_{xy}^z from PAW spin and full velocity matrices on the 12×12 VASP mesh, shown in the Cartesian first BZ. Periodic bilinear interpolation is used only for the color map. (b) Mesh refinement retaining 48 bands from 64-band VASP sources. (c) Retained-band studies from 48- and 80-band VASP sources at a fixed 6×6 mesh. The numerical integrals always use the original k points and weights.

Retaining 48 states from 64-band VASP calculations, the 6×6 , 12×12 and 18×18 meshes give $\sigma_{xy}^z = 1.382558$, 0.690723 and 0.647020 in $(\hbar/e)(e^2/h)$. The last change is 0.04370 , or 6.75% of the 18×18 value. **These meshes do not establish a converged spin Hall conductivity.** The main difficulty is the narrow response near Γ : its local value stays near $245.9 \text{ \AA}^2$, but its weighted contribution falls from 1.286413 to 0.321603 and 0.142934 . The nearest sampled radius falls from 0.276 to 0.138 and $0.092 \text{ \AA}^{-1}$. Smoothing the plot cannot supply the missing integration resolution.

The band study probes a different error. At fixed 6×6 sampling, retaining 40, 44 and 48 bands from the 48-band source gives 1.386328 , 1.384098 and 1.382555 . Increasing the actual VASP source to 64 bands while retaining the same first 48 changes the response by only 2.64×10^{-6} . The uppermost new empty states fail a Kramers-pair accuracy check. An 80-band VASP run resolves the first 64 states: retaining 48, 56 and 64 gives 1.382558 , 1.383047 and 1.381001 , with a final change of 0.15% . Only the verified subspace is used; an electronic stopping criterion alone does not establish accuracy of every stored empty state. Neither band stability nor an integer Z_2 invariant establishes k convergence.

The 12×12 charge response is $1.82 \times 10^{-7} e^2/h$, consistent with the vanishing anomalous charge Hall effect of this nonmagnetic system. With 64 source bands and the accurate first 48 retained, that residual falls to $-2.29 \times 10^{-12} e^2/h$; the 18×18 residual is $4.91 \times 10^{-9} e^2/h$, without time-reversal averaging. The finite spin response uses a different current vertex and is allowed by time reversal. Its noninteger value is not a failed Chern-number calculation. The finite-band current-product approximation, conventional-current definition, sampling and operator scope remain explicit in the output.

The supplied actual 6×6 and 12×12 PAW matrix bundles reproduce the VASPBERRY integration without rerunning VASP. The fresh SCF density, VASP input preparation and original producer insertion routines document the earlier stages; the additional Wannier inputs belong to the separate check in Appendix B.2. Every response component was independently recomputed from the raw exported matrices; all 27 spin components agree to within 3.1×10^{-14} in the stated sheet units. A separate finite-k check of the diagonal velocity gives a maximum error of $1.74 \times 10^{-4} \text{ eV \AA}$. The latter is a producer validation on development test points, not a material convergence result.

Spin Hall reproduction guide · Numerical conventions · Validation details

Appendix B. Optional Wannier supporting validation

The calculations in this appendix require an additional VASP-derived Wannier model. External Wannier90 prepares the representation; VASPBERRY imports the supplied Hamiltonian and position matrices and evaluates the stated quantities itself. It neither generates the VASP electronic states nor performs Wannier localization. These checks support bulk topology and response verification while the main user workflow remains direct WAVECAR analysis.

B.1 Full-connection evaluation of the supplied model

As a separate supporting route, VASPBERRY interpolates the Hamiltonian and position matrices of a VASP-derived Wannier model. Its own Fourier, diagonalization and occupied-trace routines evaluate $\Omega = J0 + J1 + J2$: the basis-connection curl, the mixed connection/Hamiltonian-derivative term, and the derivative-pair term. Only occupied-to-empty denominators occur; internal degeneracies are allowed. The formulation follows Wang et al. and Lopez et al., Eq. (51). Bands and local curvature are checked against direct VASP data. The separate postw90 calculation in Wannier90 provides an independent check of VASPBERRY's evaluation of the same finite model.

B.2 Bi: ideal-edge support for the native bulk Z_2 result

The auxiliary model uses 16 s/p spinor orbitals and a 6×6 VASP training mesh; independent 12×12 and 18×18 VASP comparisons test its dispersion. Finite strips of 20 and 40 cells test the width dependence.

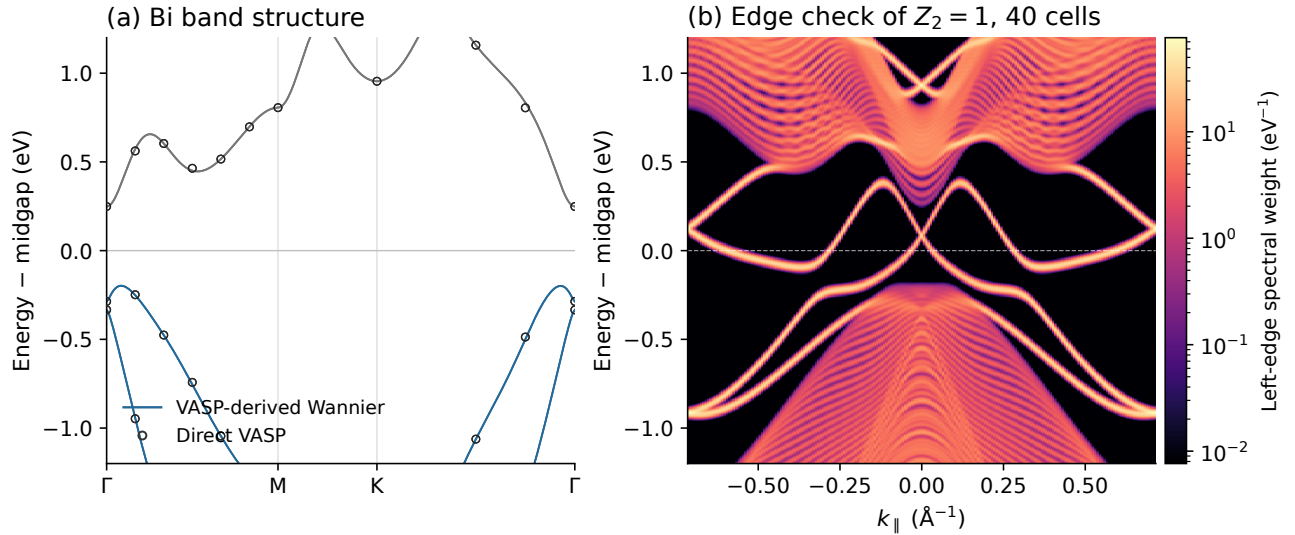


Figure 15. (a) Bi band structure from a VASP-derived 16-orbital Wannier Hamiltonian along Γ -M-K- Γ . Circles show direct VASP energies on the independent 12×12 mesh; all ten occupied bands are retained in the model. (b) Spectral weight in the first two cells of a 40-cell strip, periodic along the first lattice vector and open along the second. All eigenstates are summed before applying a 10 meV Gaussian display broadening. Energy zero is the midpoint of the sampled VASP bulk gap. The edge branches connect the valence and conduction manifolds across that gap.

The separation of the central Kramers doublets at Γ falls from 7.354 meV for a 20-cell strip to 0.03046 meV for 40 cells. Three crossings per edge occur over the positive half of the one-dimensional BZ at each of three test energies inside the bulk gap. This odd, edge-localized connectivity corroborates $Z_2 = 1$ and the time-reversal-protected boundary modes expected for this phase. The edge calculation is a supporting check of the bulk topology in Figure 7.

The VASP-derived interpolation was checked on separate 12×12 and 18×18 meshes: its sampled gaps differ from VASP by at most 2.426 meV and bands 9–12 by at most 33.23 meV. Localization stopped before its requested spread tolerance. The strip represents an ideal truncation of bulk hoppings, with no edge relaxation or device contacts. The Bi reproduction guide retains the complete interpolation, symmetry and strip-width checks.

B.3 MnBi₂Te₄: dense integration of a supplied finite model

Section 3.6 gives the complete occupied-wavefunction invariant $C = -1$ and the unresolved coarse optical integral. This optional calculation tests the corresponding Hall expectation with a VASP-derived Wannier representation.

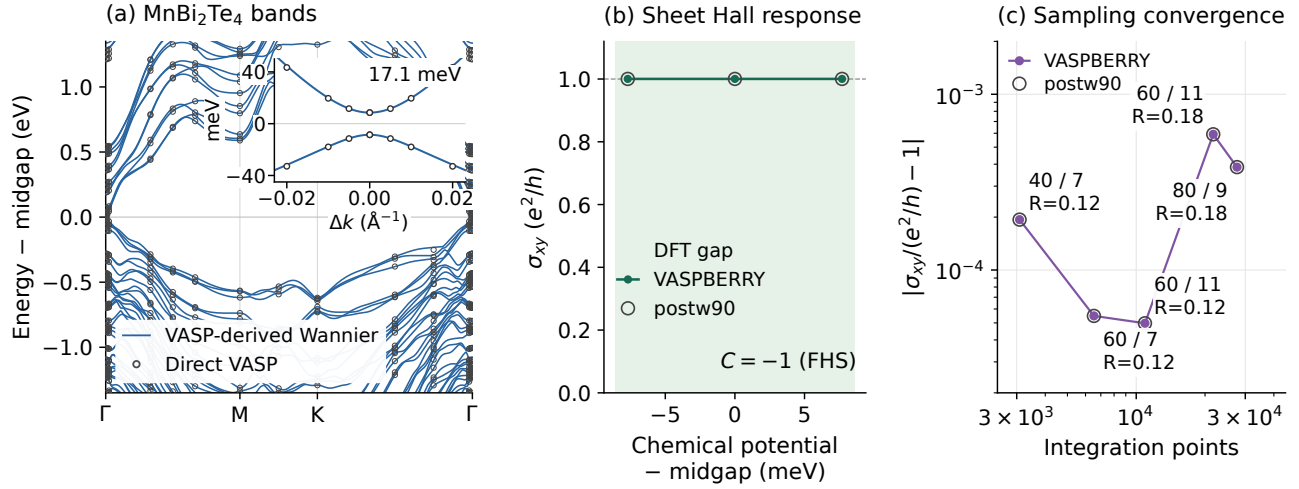


Figure 16. (a) VASP-derived Wannier band structure along Γ -M-K- Γ , with direct VASP checks; the inset resolves the K- Γ -M gap near Γ . (b) VASPBERRY full-connection sheet Hall response at three chemical potentials inside the gap, at zero temperature; open markers show the independent postw90 comparison. The dashed line is the topological expectation. (c) Deviation $|\sigma_{xy}/(e^2/h) - 1|$ under integration refinement. Labels give the base mesh, local subdivision and Γ -region radius in $\text{\AA}^{-1}$. Both solvers use the same model and weights; their agreement is distinct from mesh convergence.

For dense integration, VASPBERRY's `wannier-hall` evaluates all J0/J1/J2 terms from the supplied VASP-derived Hamiltonian and position matrices. `wannier-bands` evaluates the plotted interpolation of the same VASP-derived electronic structure. The model contains 138 orbitals and **87 occupied states**; the omitted 36 deep VASP bands form a separately checked $C = 0$ bundle. Near Γ , direct VASP band edges agree within 1.26 meV and local curvature within 1.44%; the latter comparison retains the full-123 versus model-87 subspace difference. The maximum band-edge difference across all twelve validation locations is 18.14 meV.

VASPBERRY's final 27,600-point result is $\sigma_{xy} = 1.000384978 e^2/h$ at all three gap chemical potentials. It uses an 80×80 base mesh and 9×9 subdivisions in the $0.18 \text{ \AA}^{-1}$ Γ region. The last refinement changes the result by $2.07 \times 10^{-4} e^2/h$, below the stated $10^{-3} e^2/h$ comparison criterion. The nonmonotonic region-expansion control is retained; no integer rounding is applied. Independent postw90 totals and all three component decompositions agree within their original output precision. A separate 160×160 energy scan places all three chemical potentials inside the sampled model gap.

This validates VASPBERRY's full-connection evaluation of a fixed finite model. The source 6×6 mesh, structure and basis still require convergence for quantitative material predictions; localization stopped before its spread tolerance. The material tutorial provides the actual inputs, VASPBERRY commands and outputs. The independent external reference is retained separately. This input route requires both Wannier operators; it does not infer missing PAW/nonlocal/SOC velocity terms from a WAVECAR alone.

References

1. T. Fukui, Y. Hatsugai and H. Suzuki, *Chern Numbers in Discretized Brillouin Zone: Efficient Method of Computing (Spin) Hall Conductances*, J. Phys. Soc. Jpn. **74**, 1674–1677 (2005).
2. D. Xiao, M.-C. Chang and Q. Niu, *Berry Phase Effects on Electronic Properties*, Rev. Mod. Phys. **82**, 1959–2007 (2010).
3. D. Xiao, G.-B. Liu, W. Feng, X. Xu and W. Yao, *Coupled Spin and Valley Physics in Monolayers of MoS₂ and Other Group-VI Dichalcogenides*, Phys. Rev. Lett. **108**, 196802 (2012).
4. T. Fukui and Y. Hatsugai, *Quantum Spin Hall Effect in Three Dimensional Materials: Lattice Computation of Z₂ Topological Invariants and Its Application to Bi and Sb*, J. Phys. Soc. Jpn. **76**, 053702 (2007).
5. X. Wang, J. R. Yates, I. Souza and D. Vanderbilt, *Ab initio calculation of the anomalous Hall conductivity by Wannier interpolation*, Phys. Rev. B **74**, 195118 (2006).
6. M. Gajdoš et al., *Linear optical properties in the projector-augmented wave methodology*, Phys. Rev. B **73**, 045112 (2006).
7. M. M. Otrokov et al., *Unique Thickness-Dependent Properties of the van der Waals Interlayer Antiferromagnet MnBi₂Te₄ Films*, Phys. Rev. Lett. **122**, 107202 (2019).
8. J.-Q. Yan et al., *Crystal growth and magnetic structure of MnBi₂Te₄*, Phys. Rev. Materials **3**, 064202 (2019).
9. G. Pizzi et al., *Wannier90 as a community code: new features and applications*, J. Phys.: Condens. Matter **32**, 165902 (2020).
10. M. Lopez, D. Vanderbilt, T. Thonhauser and I. Souza, *Wannier-based calculation of the orbital magnetization in crystals*, Phys. Rev. B **85**, 014435 (2012).
11. J. H. Ryoo, C.-H. Park and I. Souza, *Computation of intrinsic spin Hall conductivities from first principles using maximally localized Wannier functions*, Phys. Rev. B **99**, 235113 (2019).
12. J. Qiao et al., *Calculation of intrinsic spin Hall conductivity by Wannier interpolation*, Phys. Rev. B **98**, 214402 (2018).
13. S. Murakami, *Quantum Spin Hall Effect and Enhanced Magnetic Response by Spin-Orbit Coupling*, Phys. Rev. Lett. **97**, 236805 (2006).
14. T. H. Yang et al., *Stacking-induced direct band gap in CVD-grown 1H MoS₂ bilayers*, Nature (2026).
15. E. Prodan, *Robustness of the spin-Chern number*, Phys. Rev. B **80**, 125327 (2009).